CAEN Electronic Instrumentation

Tools for Discovery

NI Nuclear Instruments

N1081B
Configurable Logic Module

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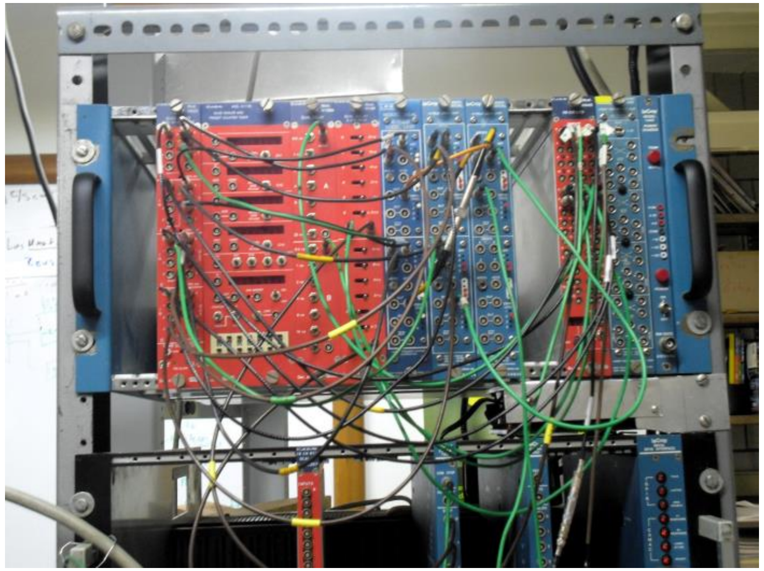
CERN – Electronic Pool – 22/6/2021

N1081B / DT1081B

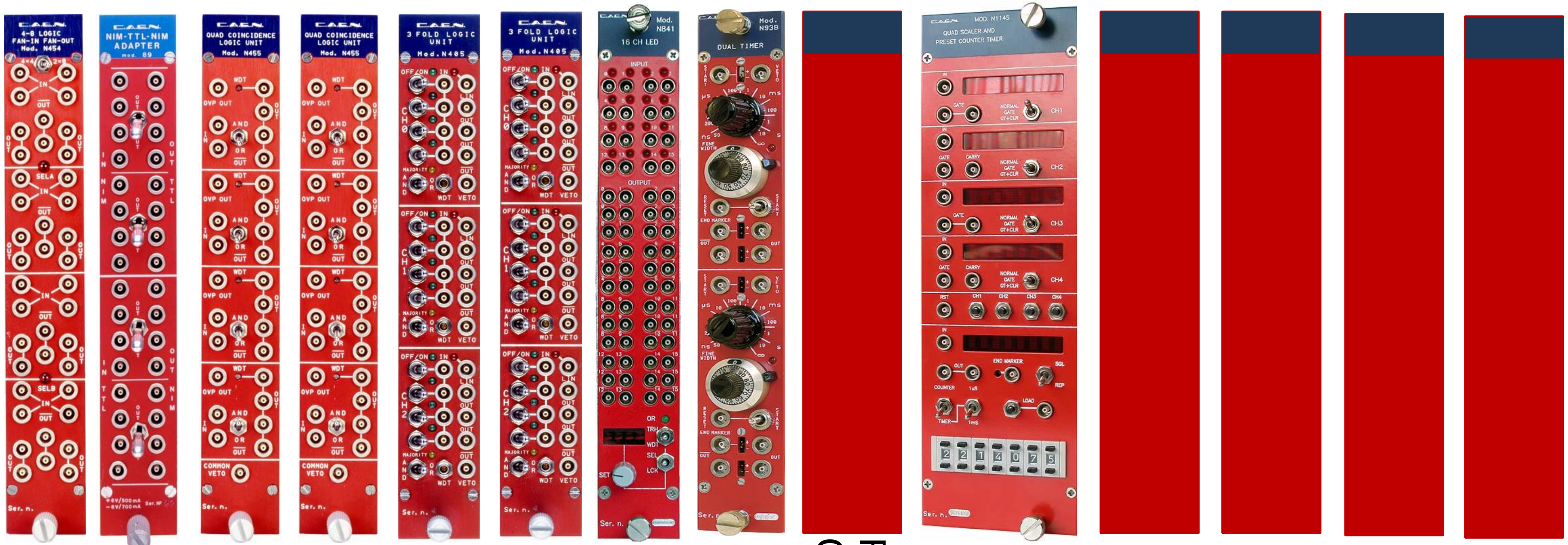
A pocketknife tool
in your laboratory!

The new way: N1081B

- **Multiple function** in a single NIM module
- **Replace** old function-specific NIM modules with a reconfigurable device
- **Reduce** the number of cables in your experiment
- **Simplify** the debug and verification process of the experimental setup
- **Remote reconfiguration** and data readout



22 Different functions in a single device



FANOUT

NIM/TTL

AND

OR

VETO

MAJORITY

DISCRIM

GATE & DELAY

PROG. COINCIDENCE

TIMER COUNTERS RATE METER

LUT

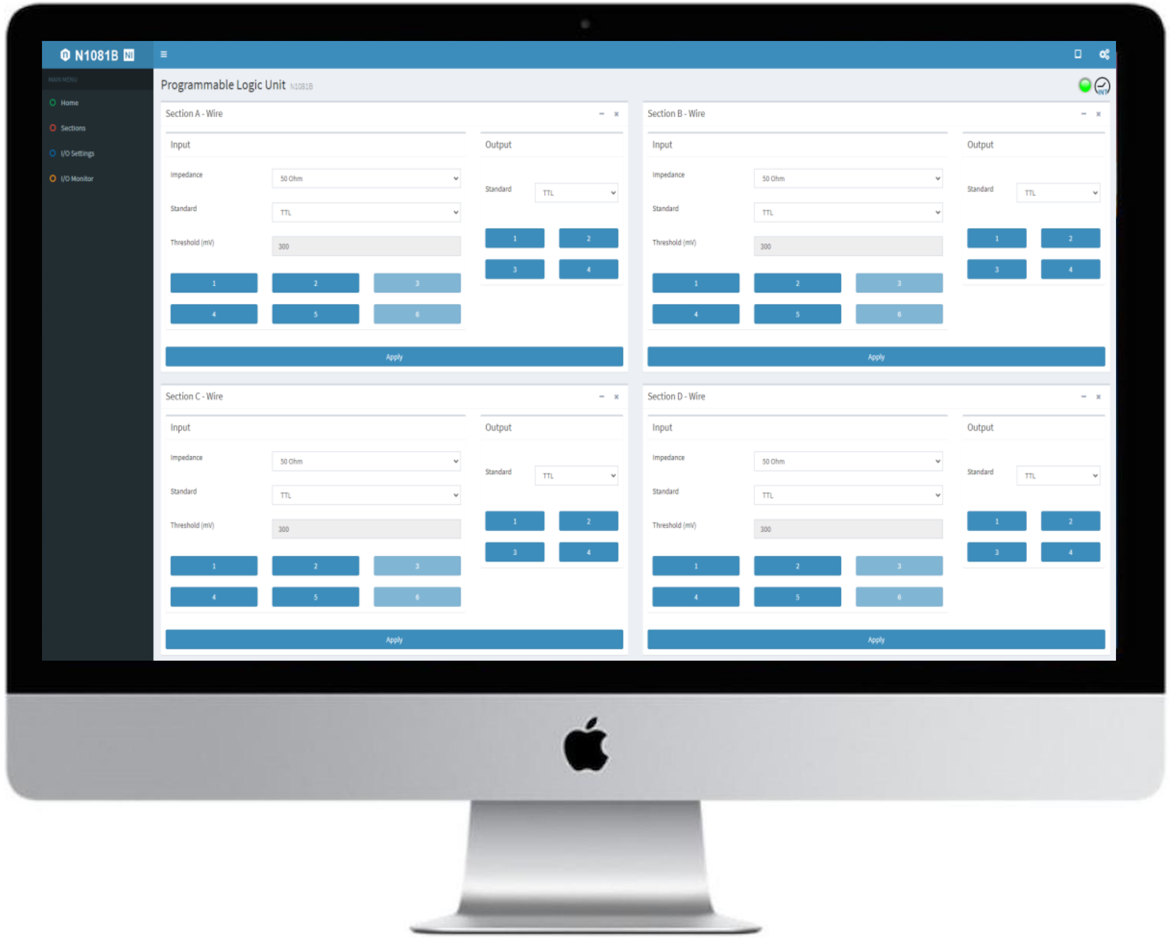
TOF TOT

TIME TAGGING

PULSE GEN

Remote Control & Monitor

CAEN x1081B can be configured remotely and data can be analyzed via web based interface



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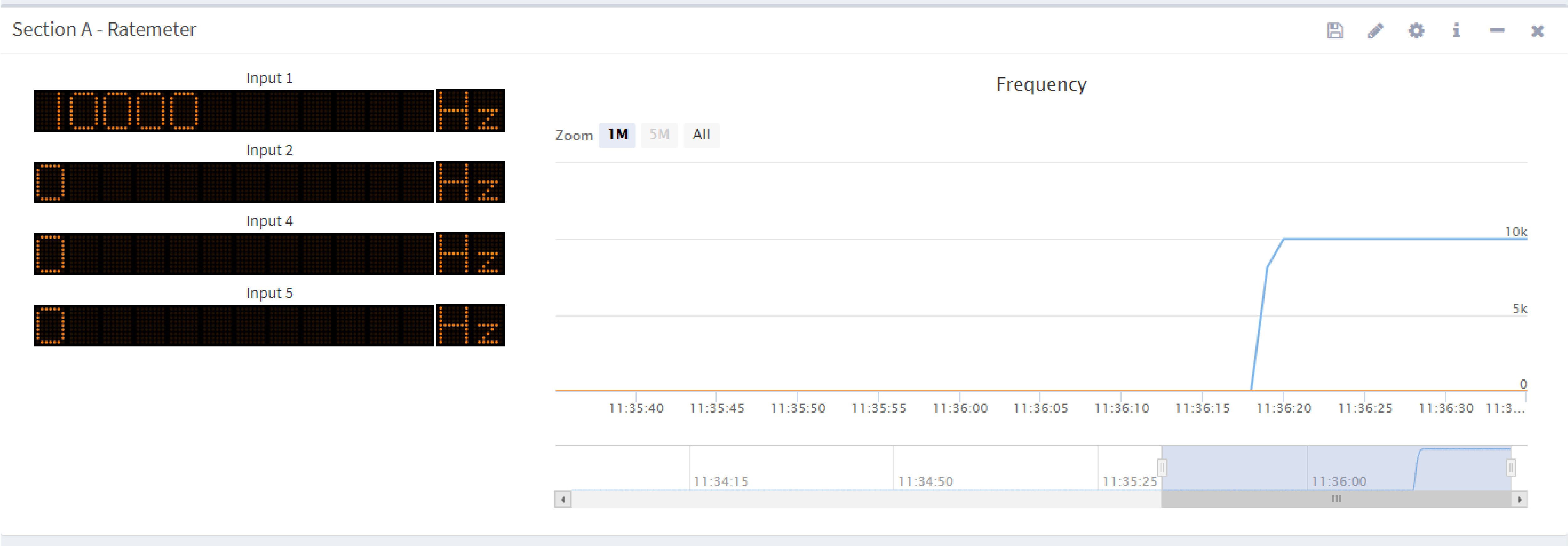

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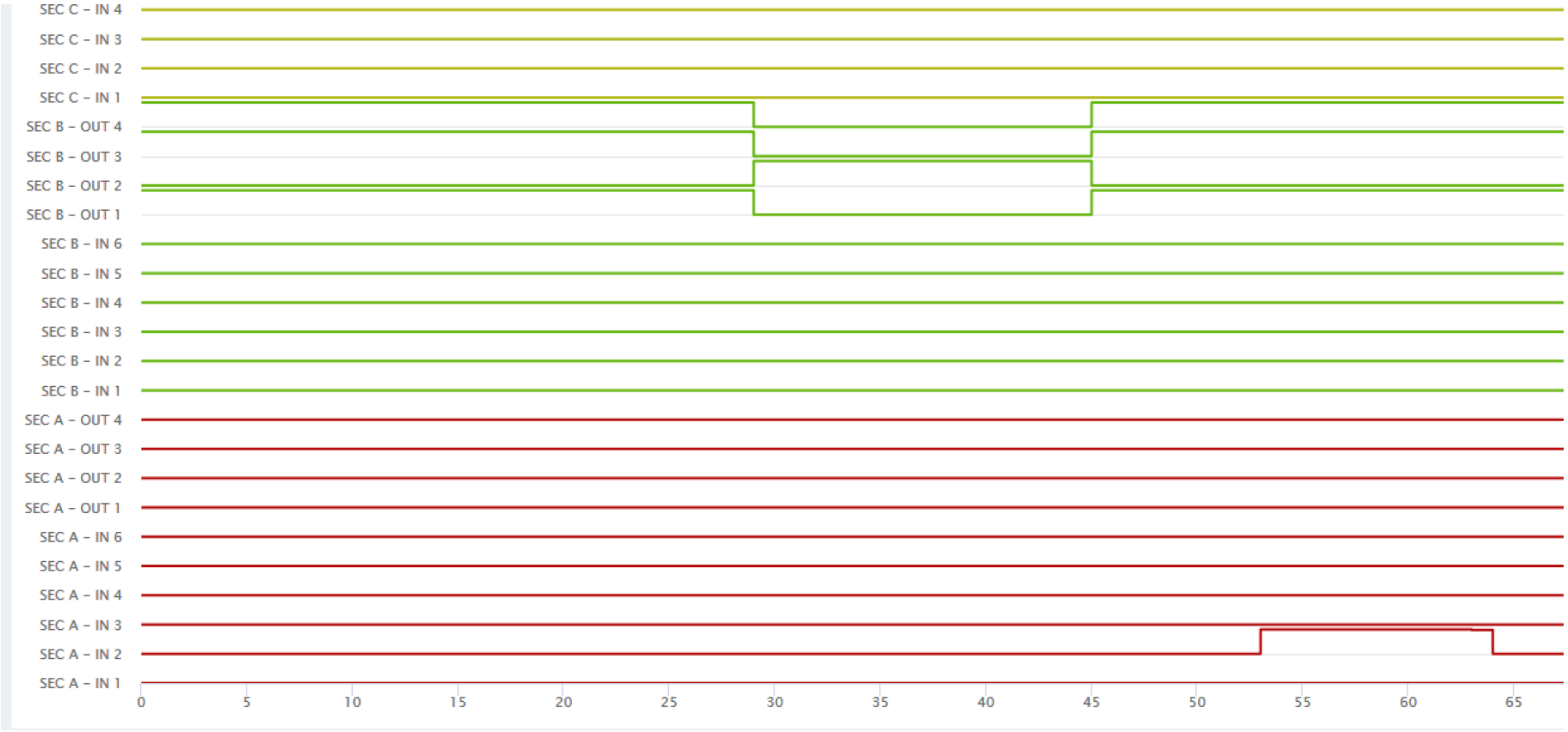

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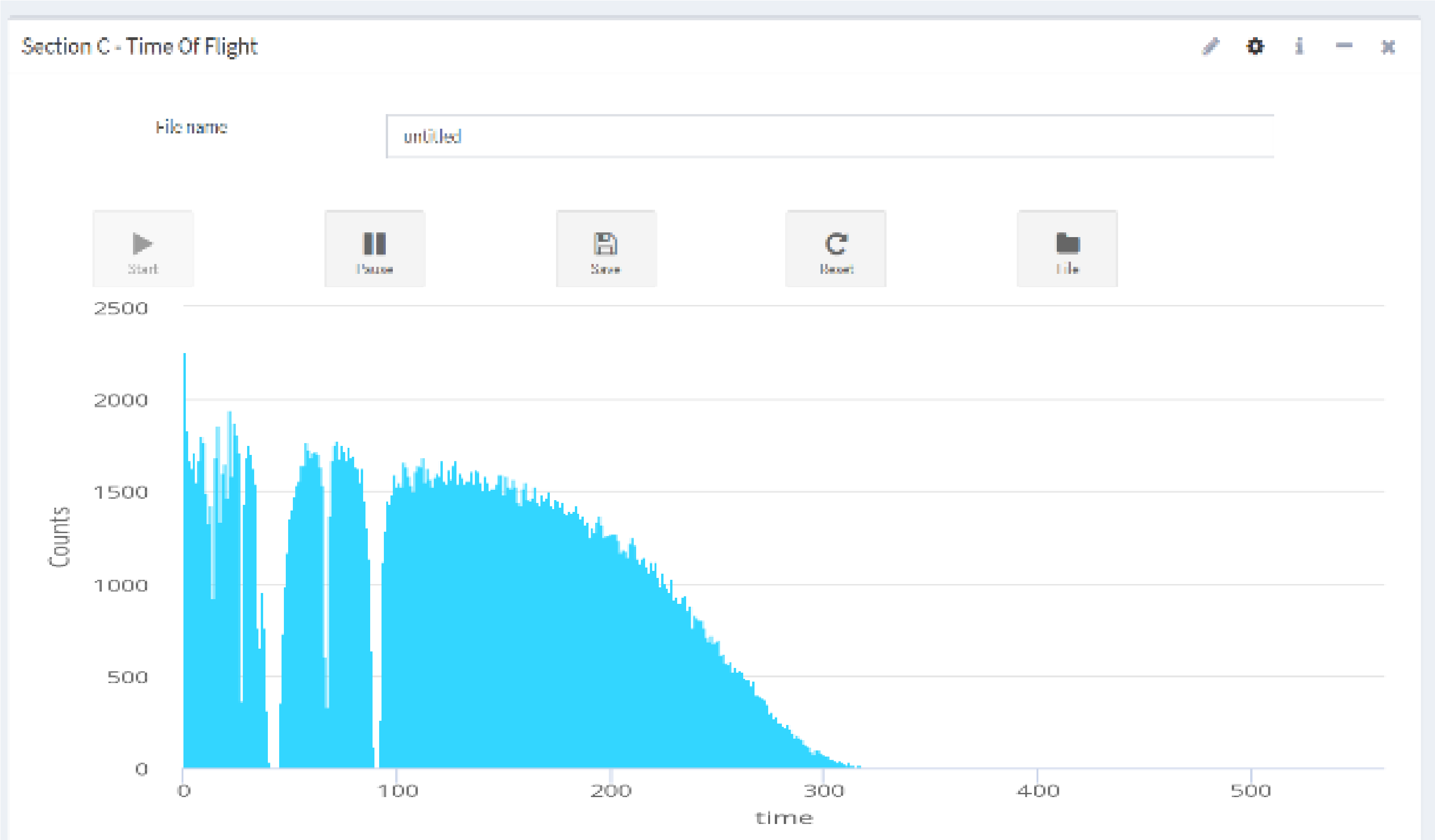


Device I/O

Touch Screen

NIM/TTL
Indicator

4-section input

4-section output

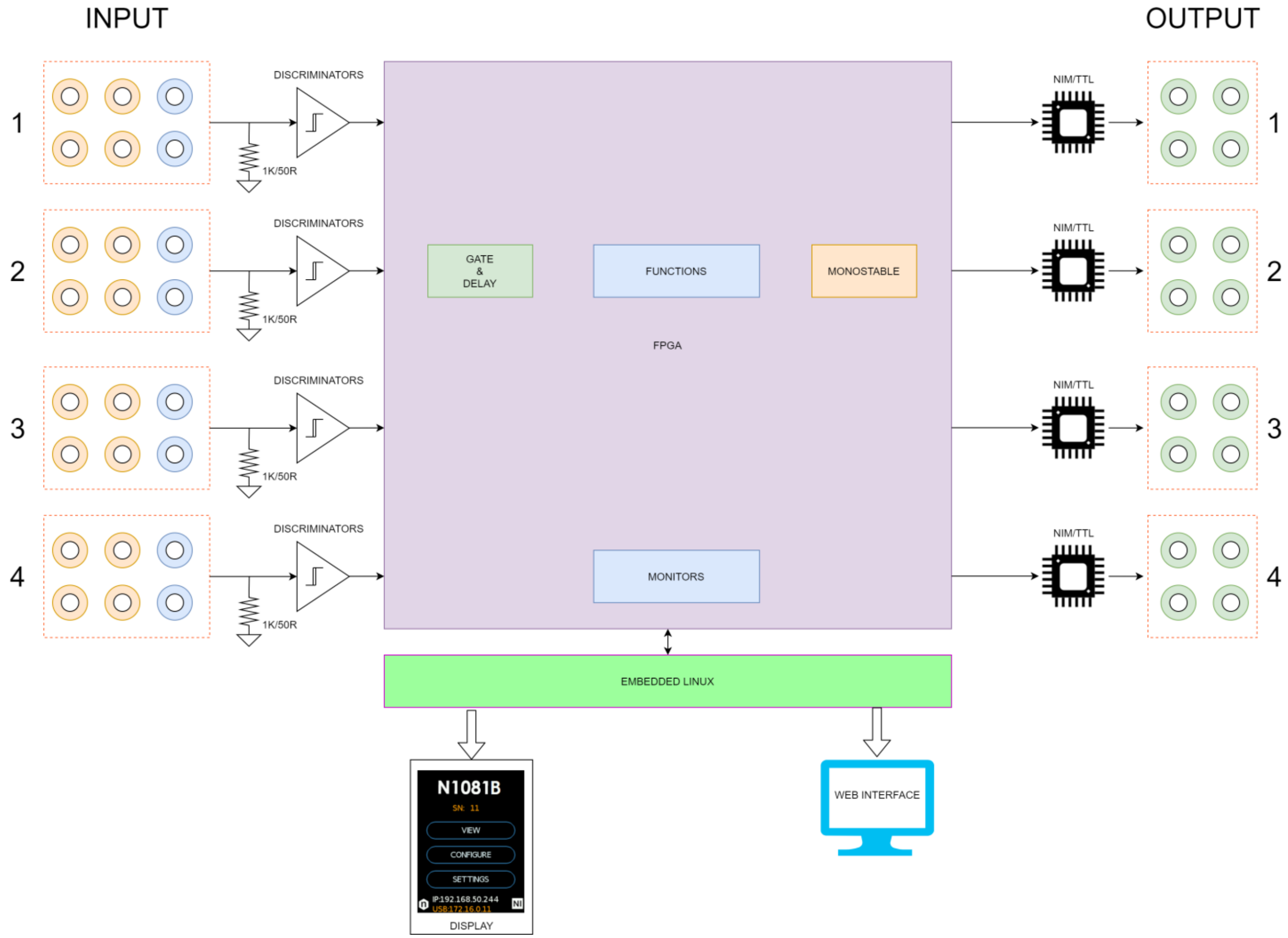
Gigabit Ethernet

USB-2.0

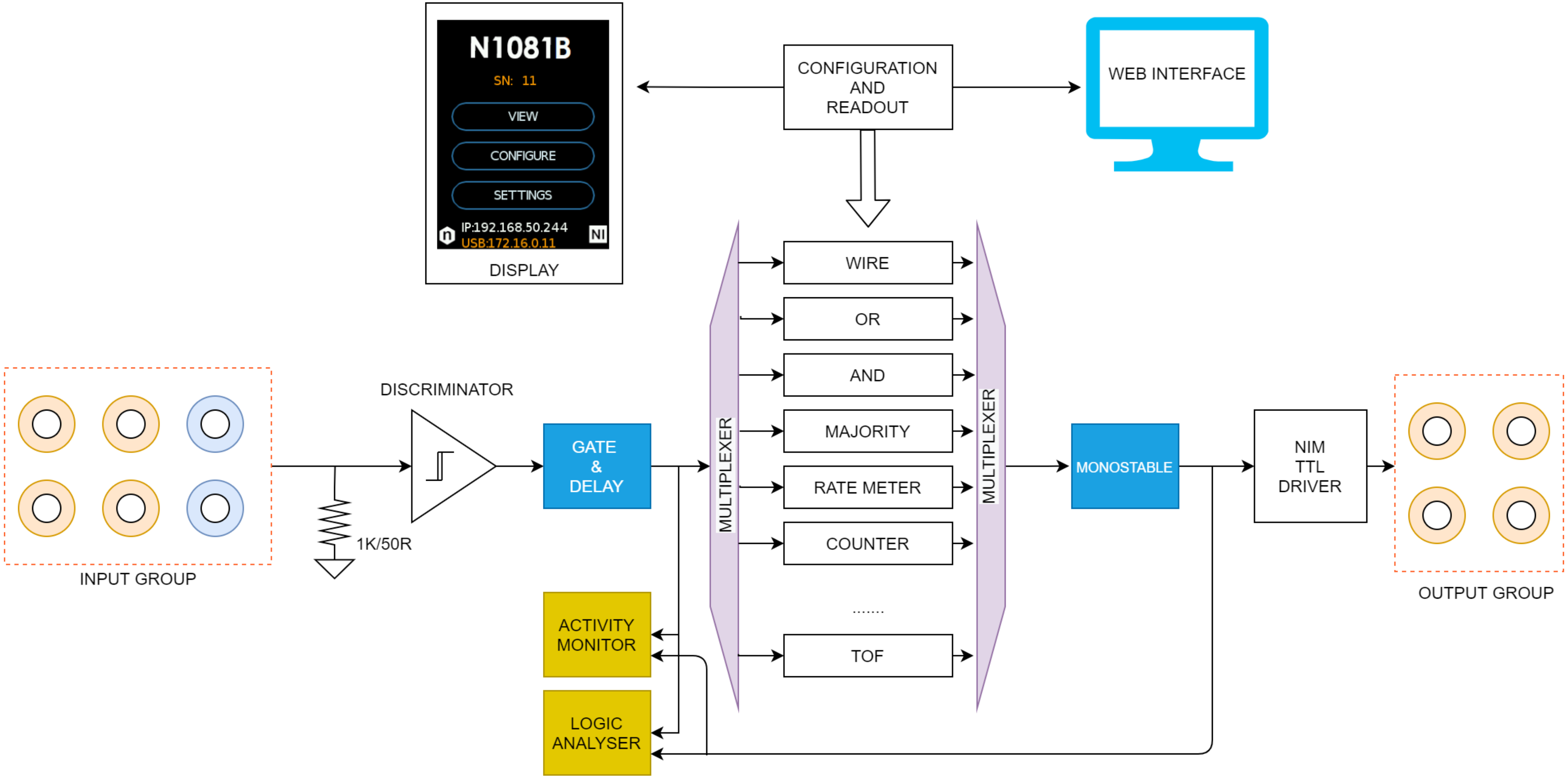
External Clock

NIM Power

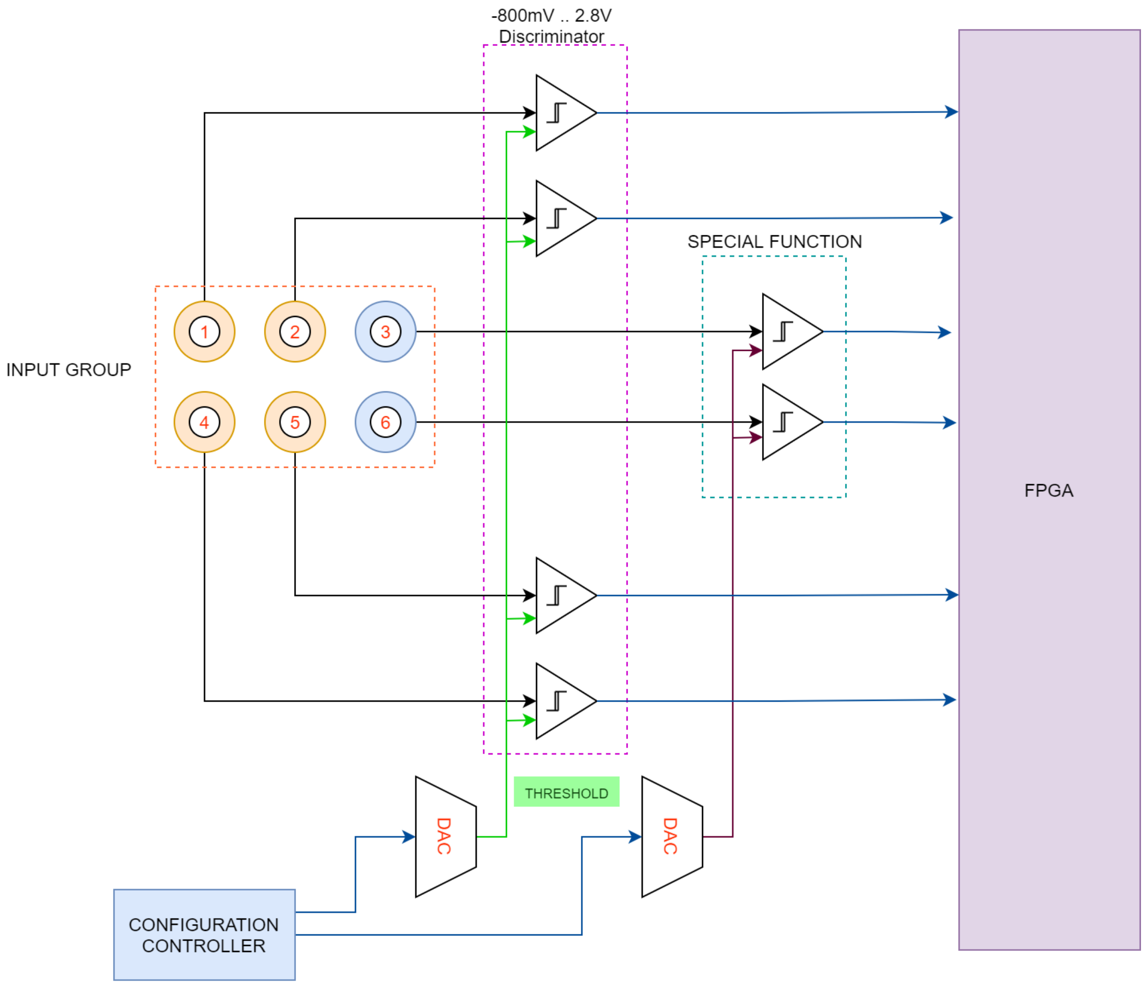
Device Architecture



Section Architecture

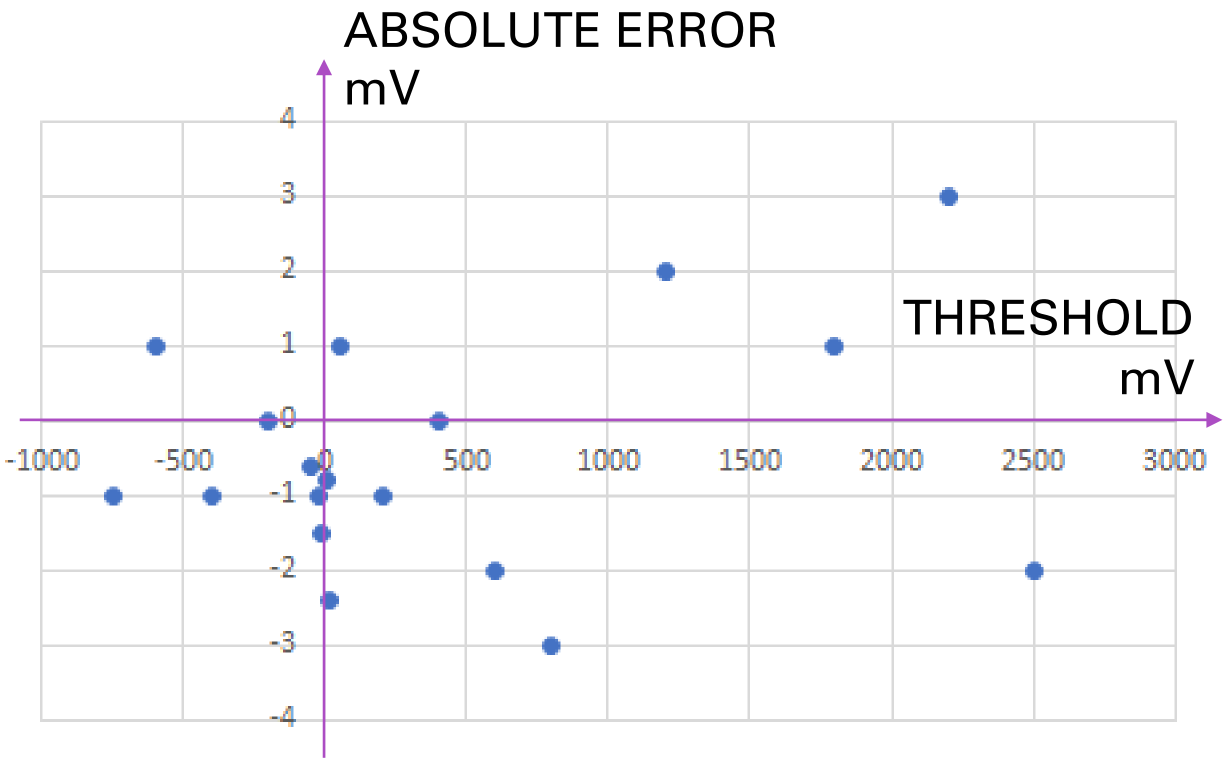


Input discriminator



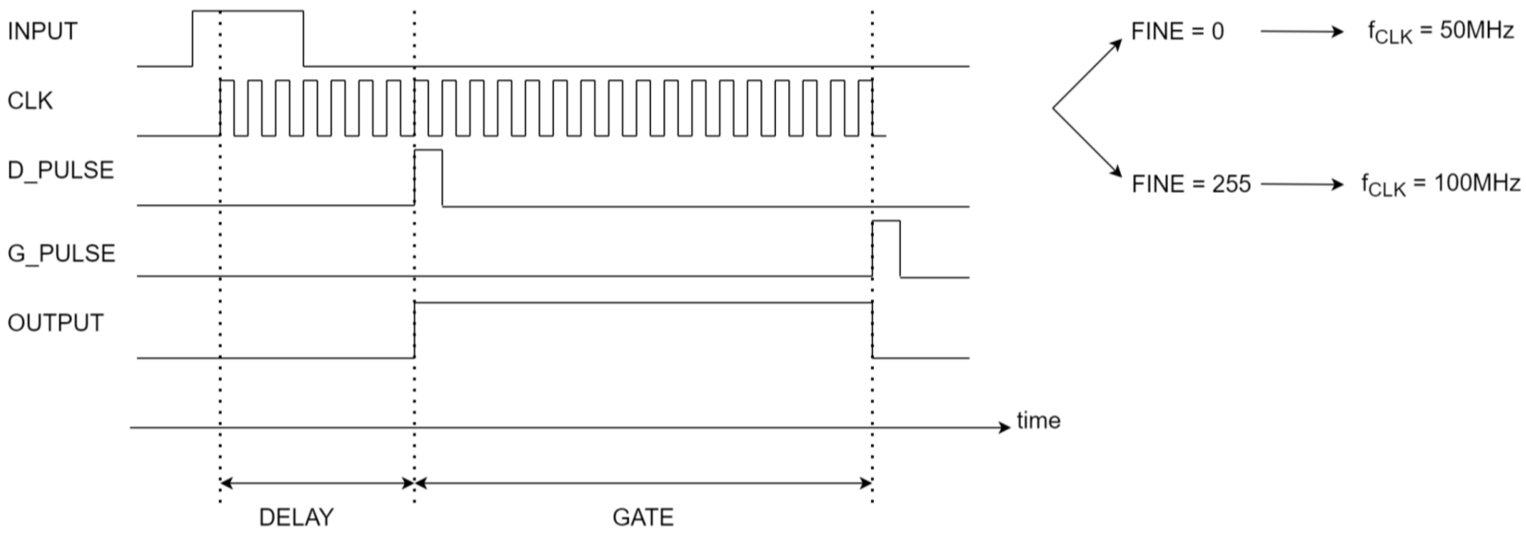
- **Bipolar discriminator** with very low selectable threshold (10mV)
- **Narrow Transition** with less than 2mV transition threshold (0 to 99%)
- **Calibrated threshold**
  Non linearity +-6.5mV +-4%
- **Extended threshold range** from -800mV to 2.5V programmable in step of 1mV
- **Fast**
  - 130 MHz with scaler
  - 100 MHz logic function NIM
  - 80 MHz logic function NIM
  - 40 MHz synchronous functions

Input discriminator



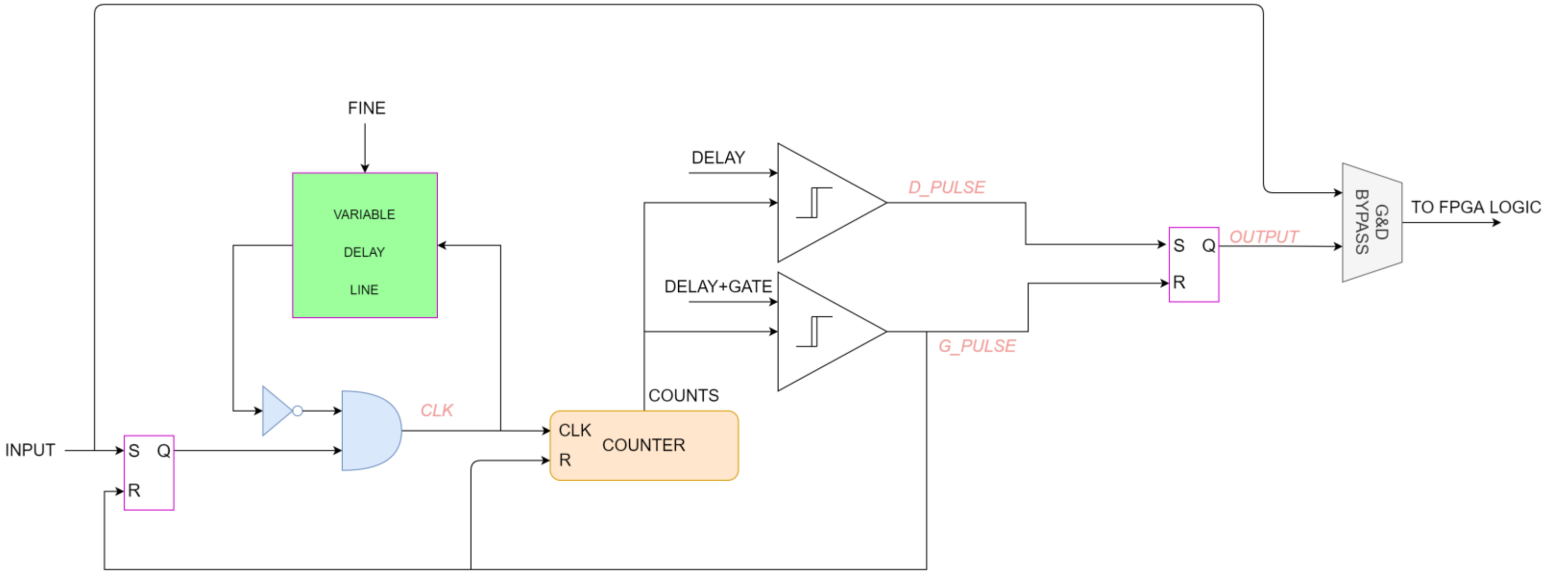
- **Bipolar discriminator** with very low selectable threshold (<10mV)
- **Narrow Transition** with less than 2mV transition threshold (0 to 99%)
- **Calibrated threshold**
  Non linearity +-6.5mV +-4%
- **Extended threshold range** from -800mV to 2.5V programmable in step of 1mV
- **Fast** (up to 120 MHz with scaler, 65 MHz with logic function and 35 MHz with timer, counters, TOT)

Input gate and delay generator



- **Arbitrary shape** input signal
- **Independent** gate and delay **for each input**
- **Programmable** from 15ns to 10us, independent delay and gate
- **Resolution** +-1%+-3ns
- **Internal ring oscillator** generate pulses synchronous with the input source minimizing the input jitter
- **Temperature compensation** automatically adjust the ring oscillator frequency to generate accurate delays

Input gate and delay generator



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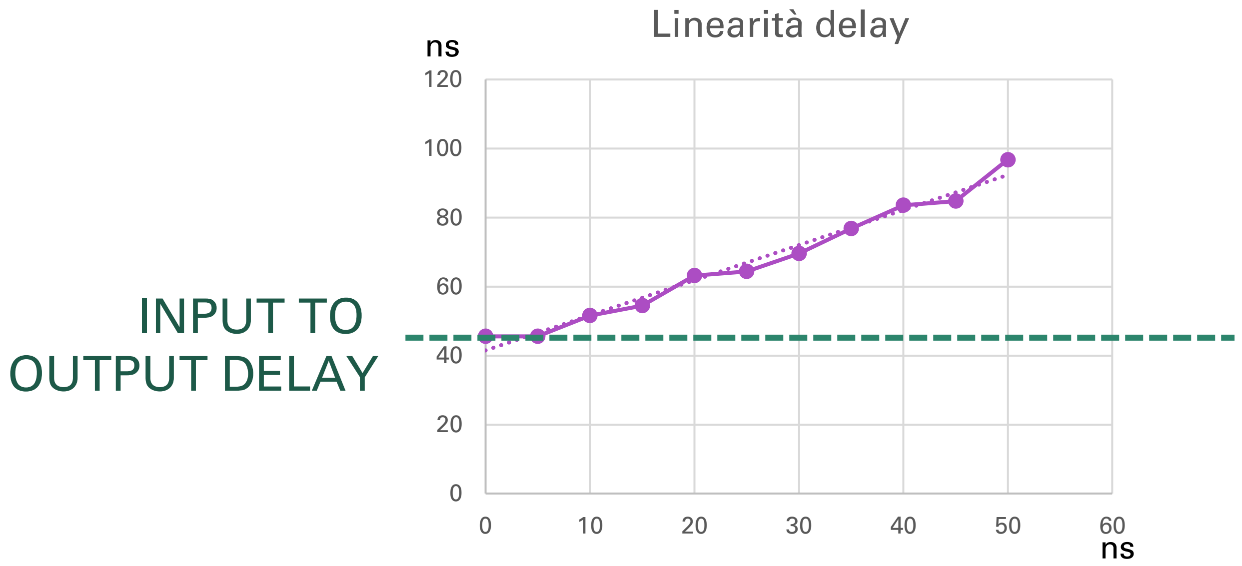

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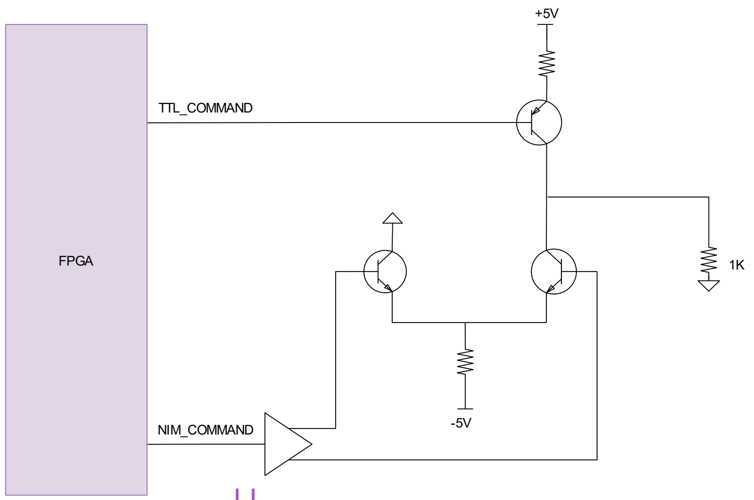
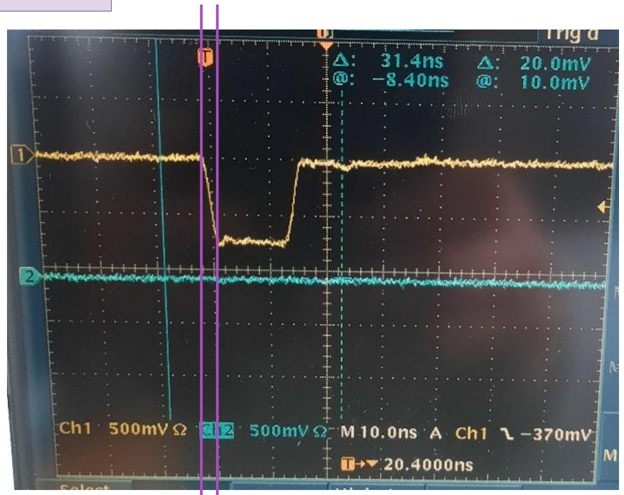
Output block diagram





1ns on NIM

- **TTL / NIM** input signal
- **Fast** output signal edge (<2ns)
- **Maximum rate**
  100MHz NIM and 80 MHz TTL

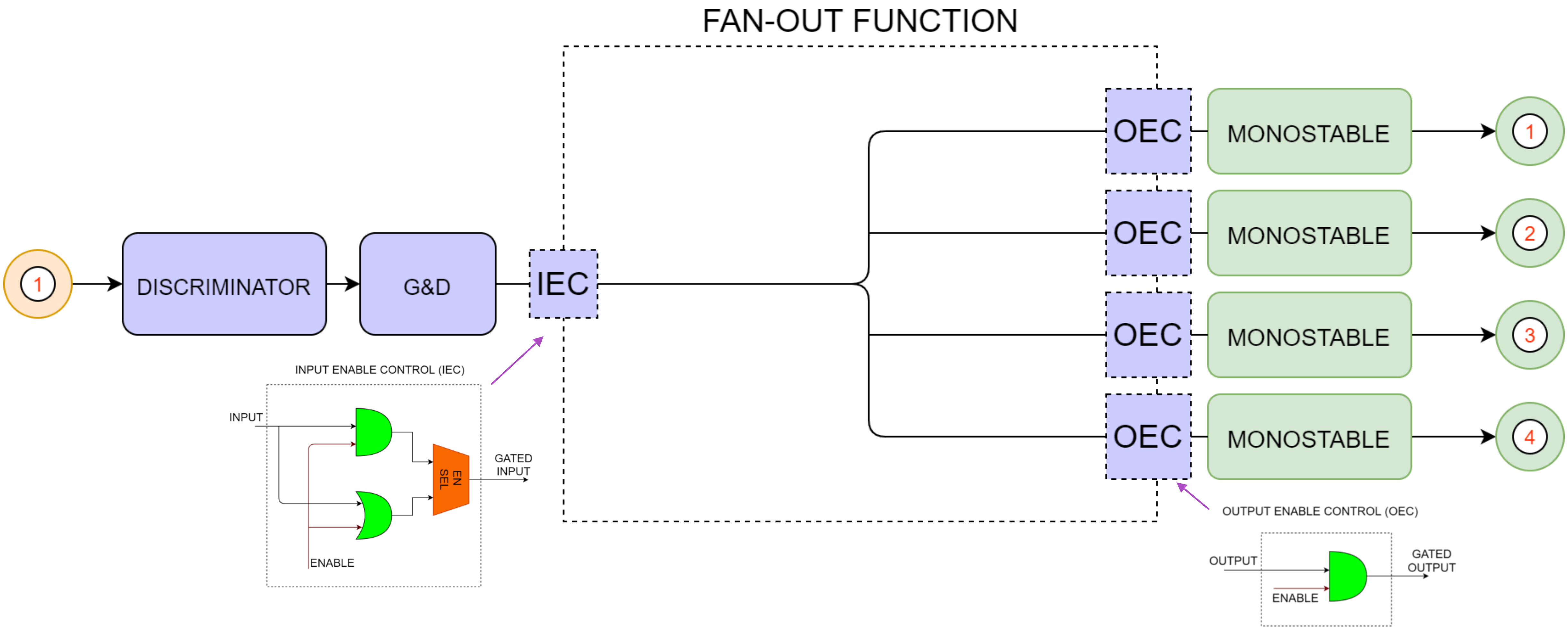
Core functions

I/O	LOGIC	COUNTER	TIMING	GENERATOR
GATE & DELAY	FANOUT	SCALER	TIME TAGGING	PULSE GENERATOR
MONOSTABLE	WIRE	COUNTER	TIME OF FLIGHT	DIGITAL GENERATOR
I/O MONITOR	AND	COUNTER TIMER	TIME OVER THRESHOLD	PATTERN GENERATOR
LOGIC ANALYSER	OR	CHRONOMETER		
	OR + VETO	RATE METER		
	VETO	RATE METER ADV.		
	MAJORITY			
	MAJORITY + VETO			
	LUT			
	COINCIDENCE			

Asyncronous logic (there is no clock sampling signal in the function and the propagation delay is proportional to the number of logic element the signal is crossing)

Syncronous to the system 100 MHz clock (the outputs of the function are updated only on rising edge of the clock signal)

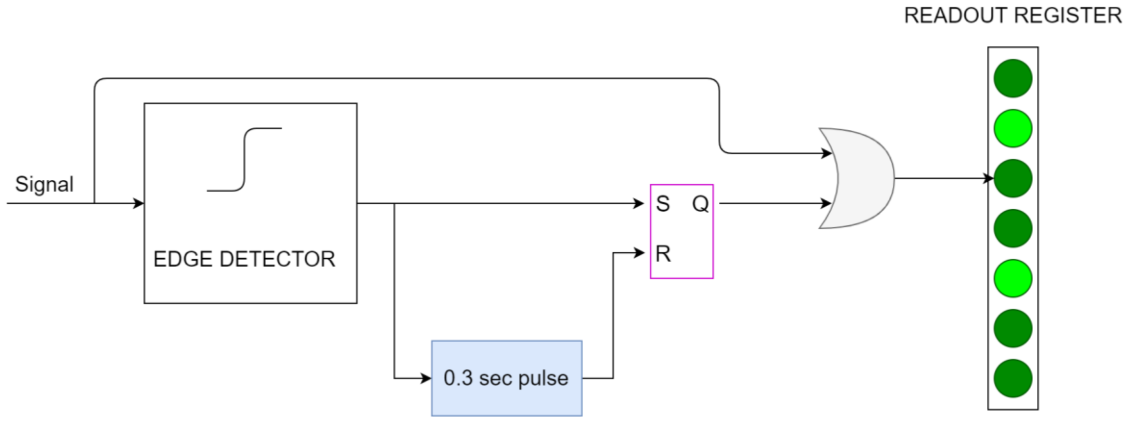
Fanout



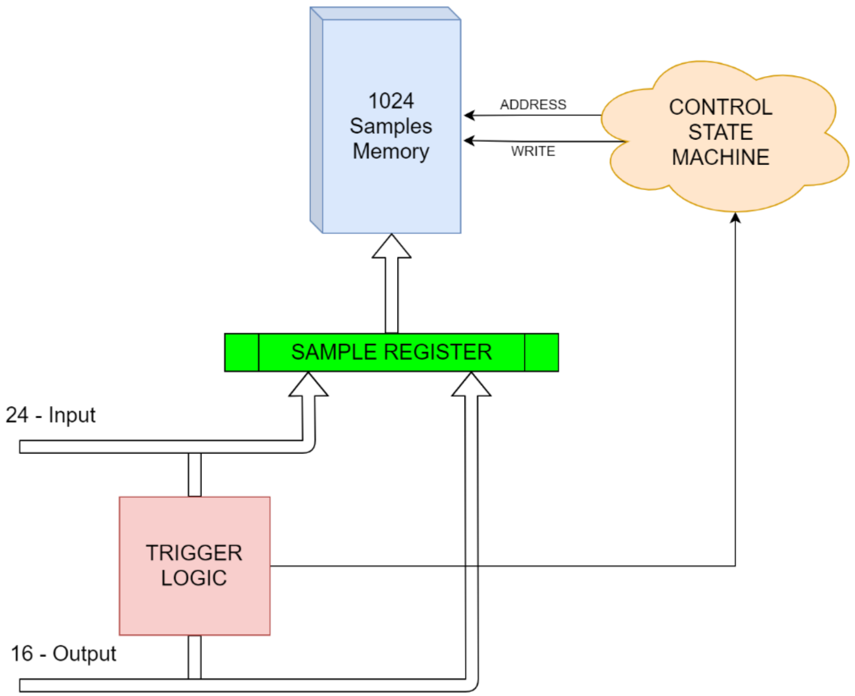
FAN-OUT FUNCTION
1
DISCRIMINATOR
G&D
IEC
OEC
MONOSTABLE
1
OEC
MONOSTABLE
2
OEC
MONOSTABLE
3
OEC
MONOSTABLE
4
INPUT ENABLE CONTROL (IEC)
INPUT
EN
SEL
GATED
INPUT
ENABLE
OUTPUT ENABLE CONTROL (OEC)
OUTPUT
ENABLE
GATED
OUTPUT

Monitor

Activity Monitor

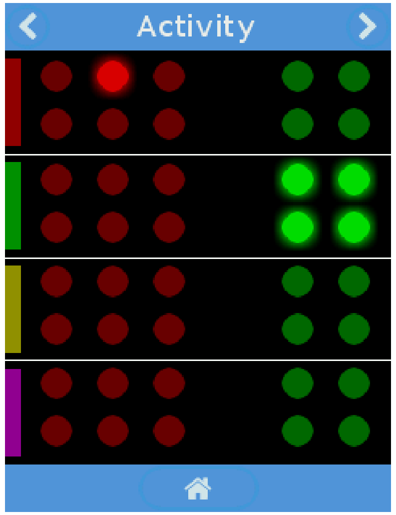


Logic Analyzer Monitor

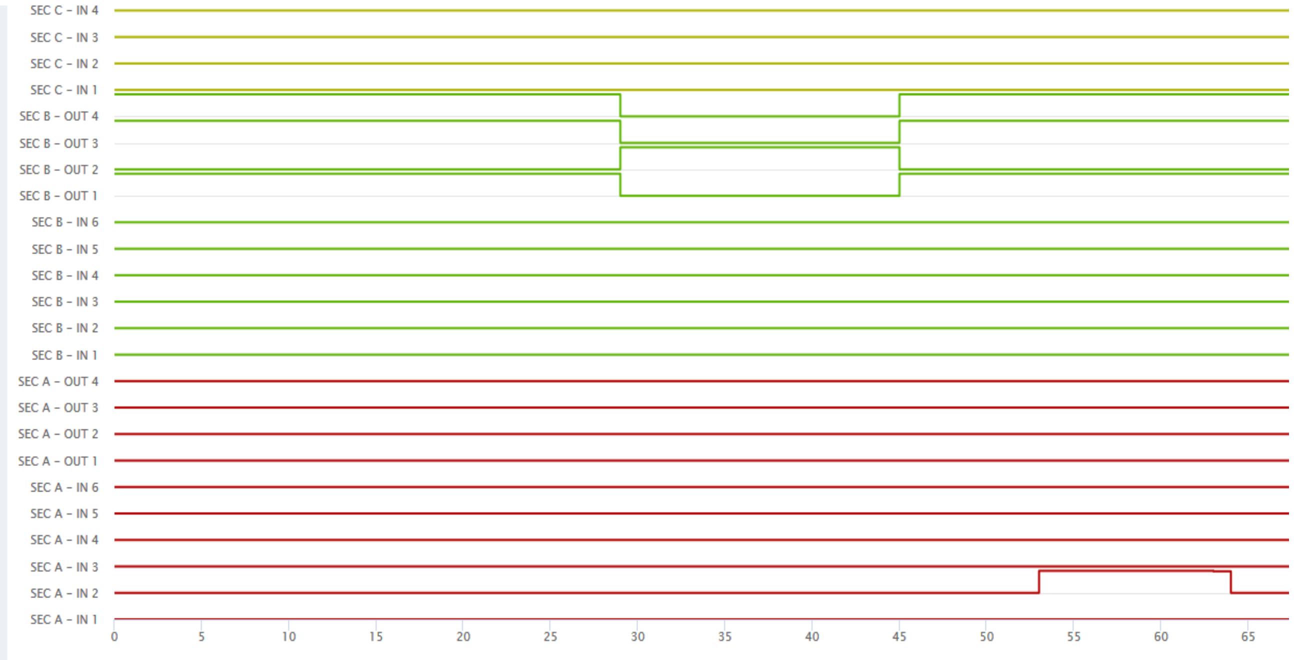


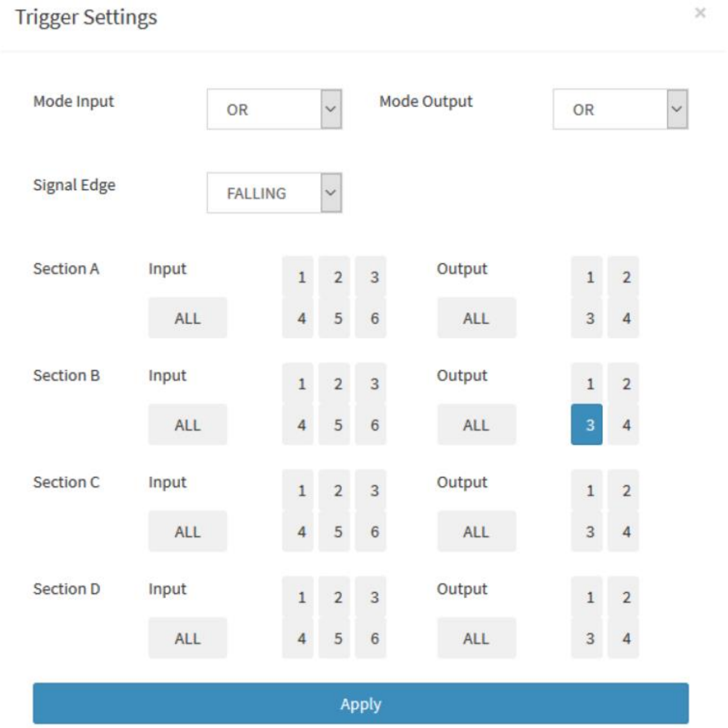
Monitor

Activity Monitor



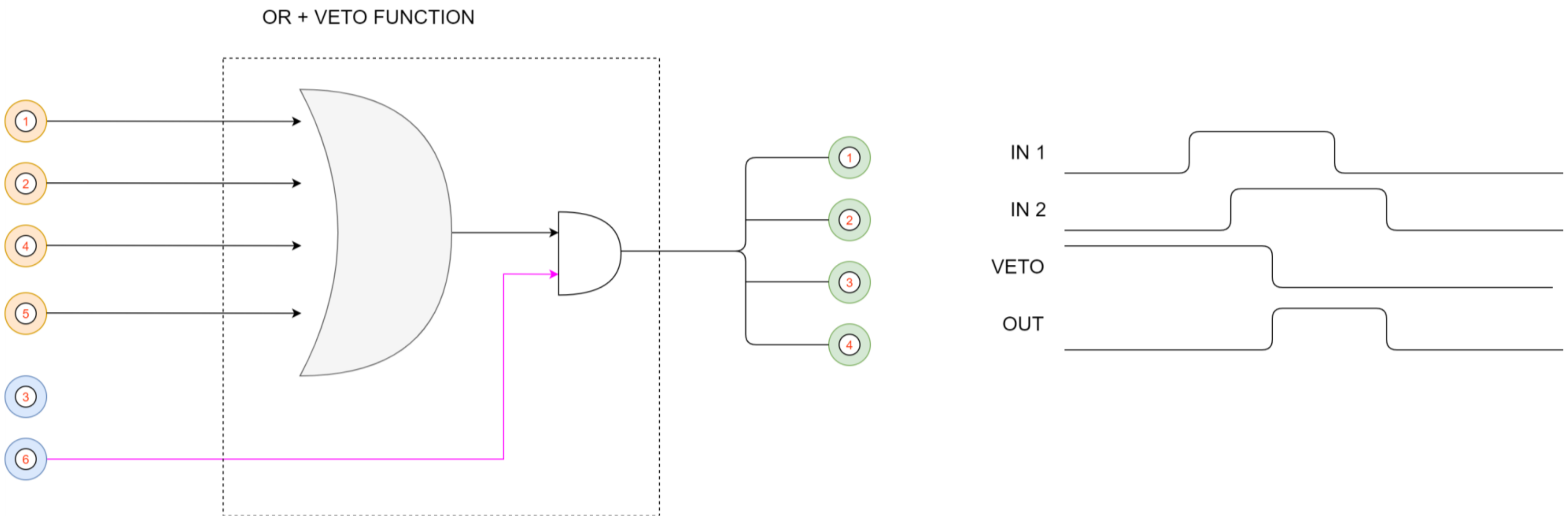
Logic Analyzer Monitor



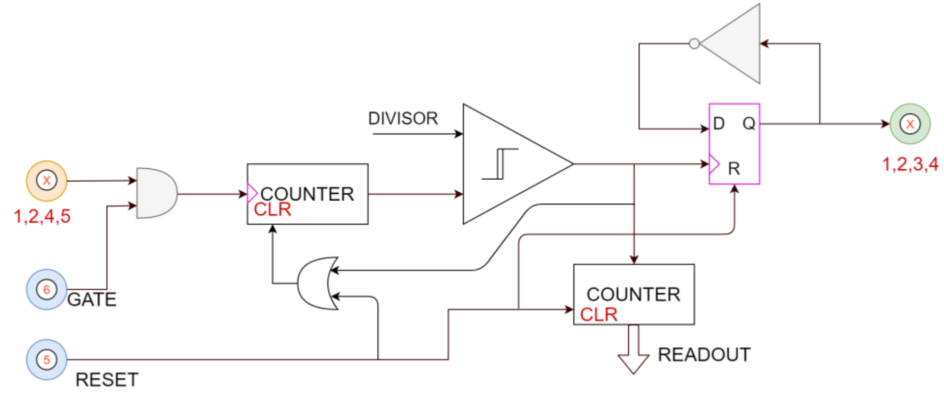


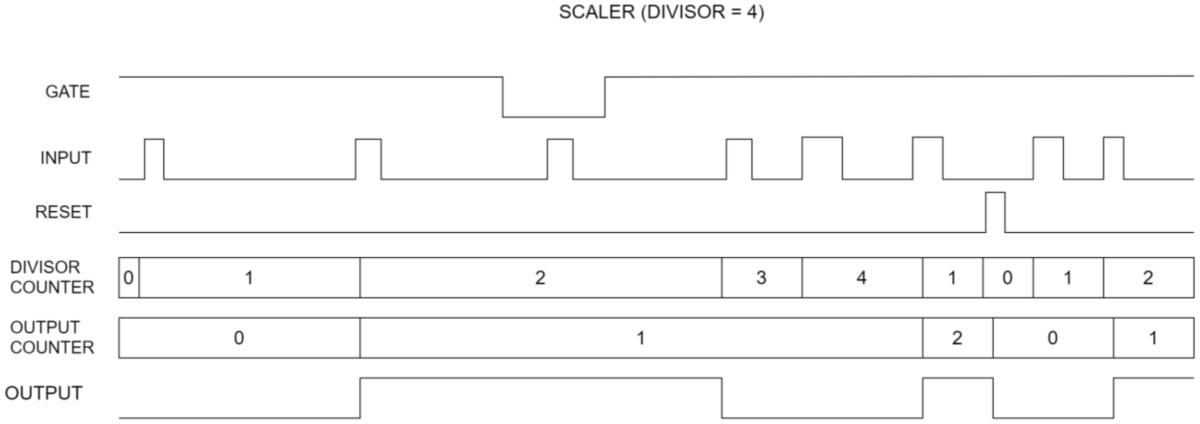
COMPLEX PROGRAMMABLE TRIGGER TO SEEK FOR RARE EVENTS

Logic functions

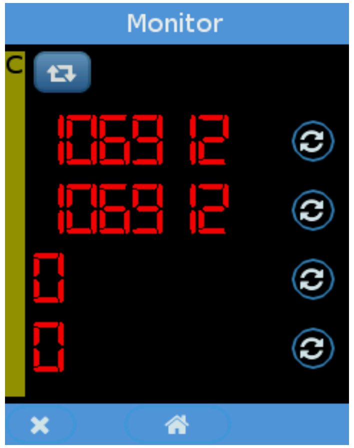


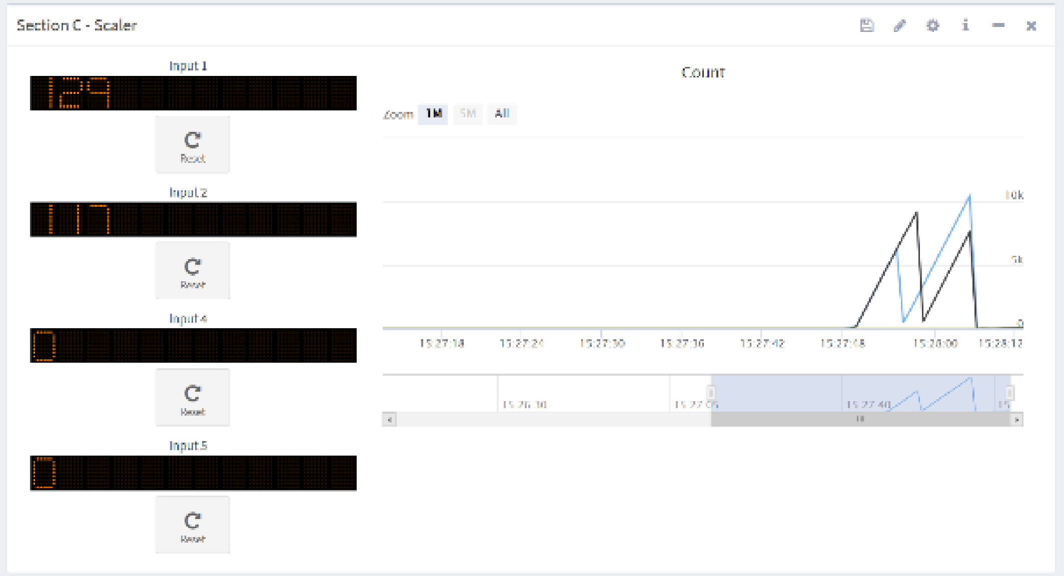
Scaler



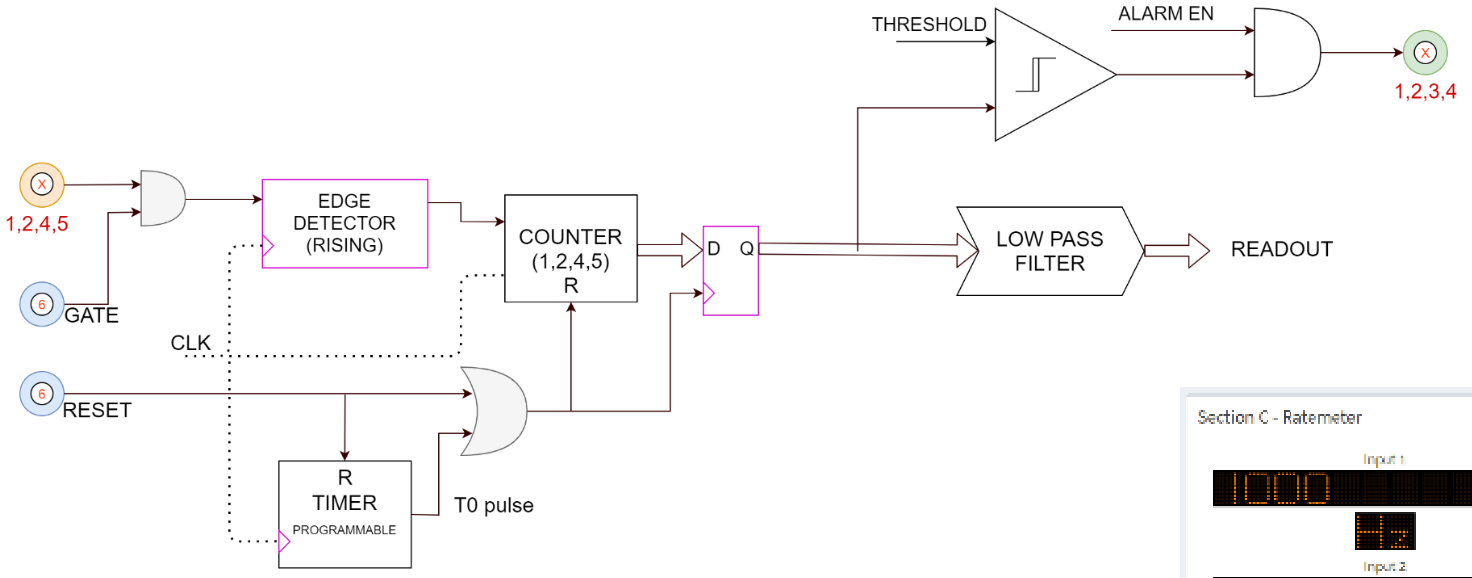


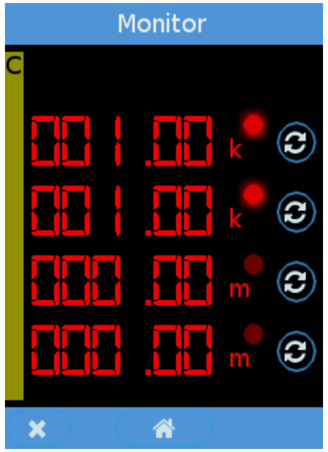
Syncronous with input signal

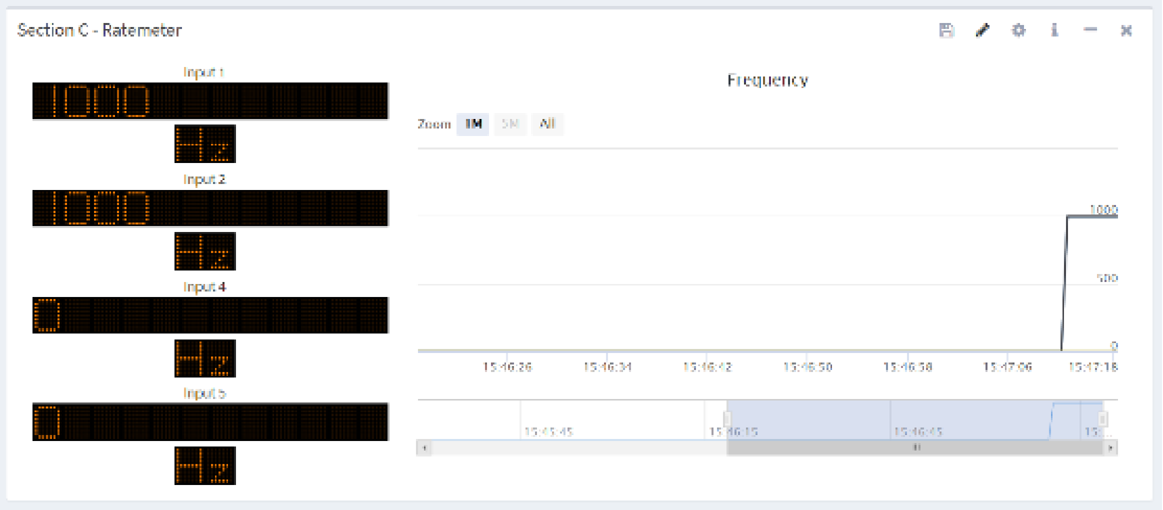




Rate Meter

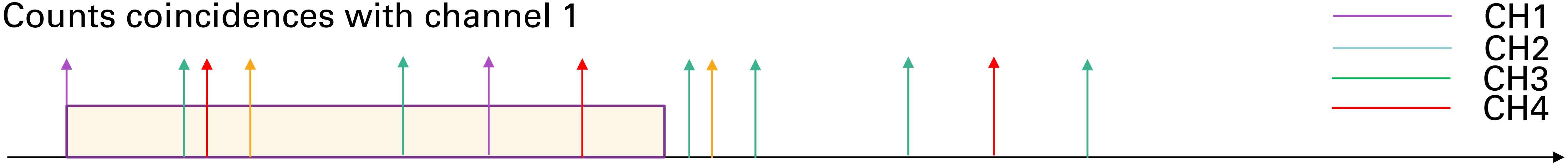

THRESHOLD
ALARM EN
1,2,3,4
1,2,4,5
EDGE DETECTOR (RISING)
COUNTER (1,2,4,5) R
D Q
LOW PASS FILTER
READOUT
GATE
CLK
RESET
R TIMER PROGRAMMABLE
T0 pulse


Monitor

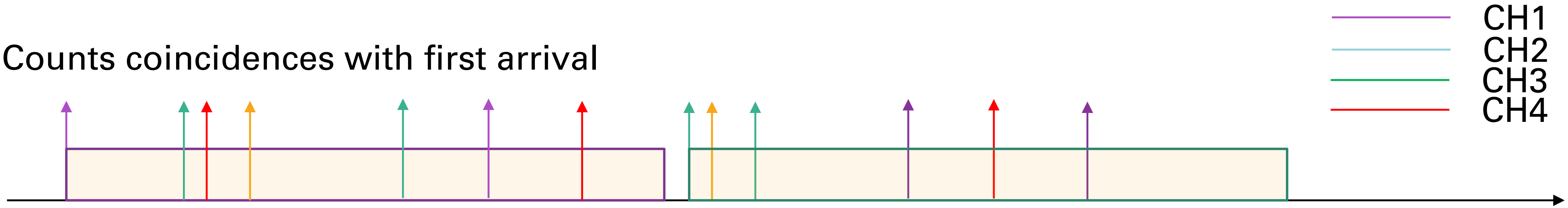

Section C - Ratemeter
Input 1
1000
Hz
Input 2
1000
Hz
Input 4
0
Hz
Input 5
0
Hz
Frequency
Zoom
1M
5M
All

Coincidence Counter

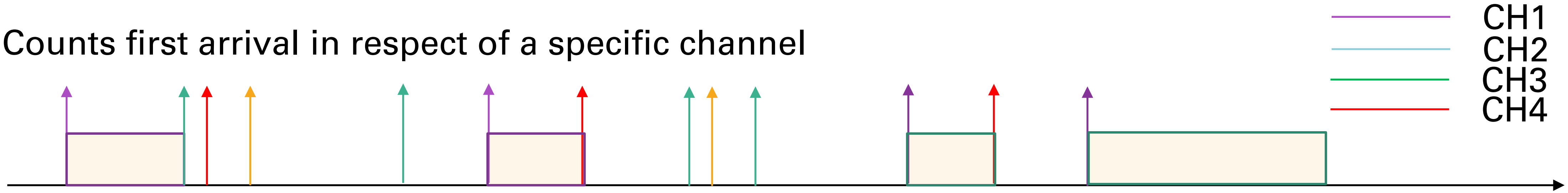
Counts coincidences with channel 1



Counts coincidences with first arrival



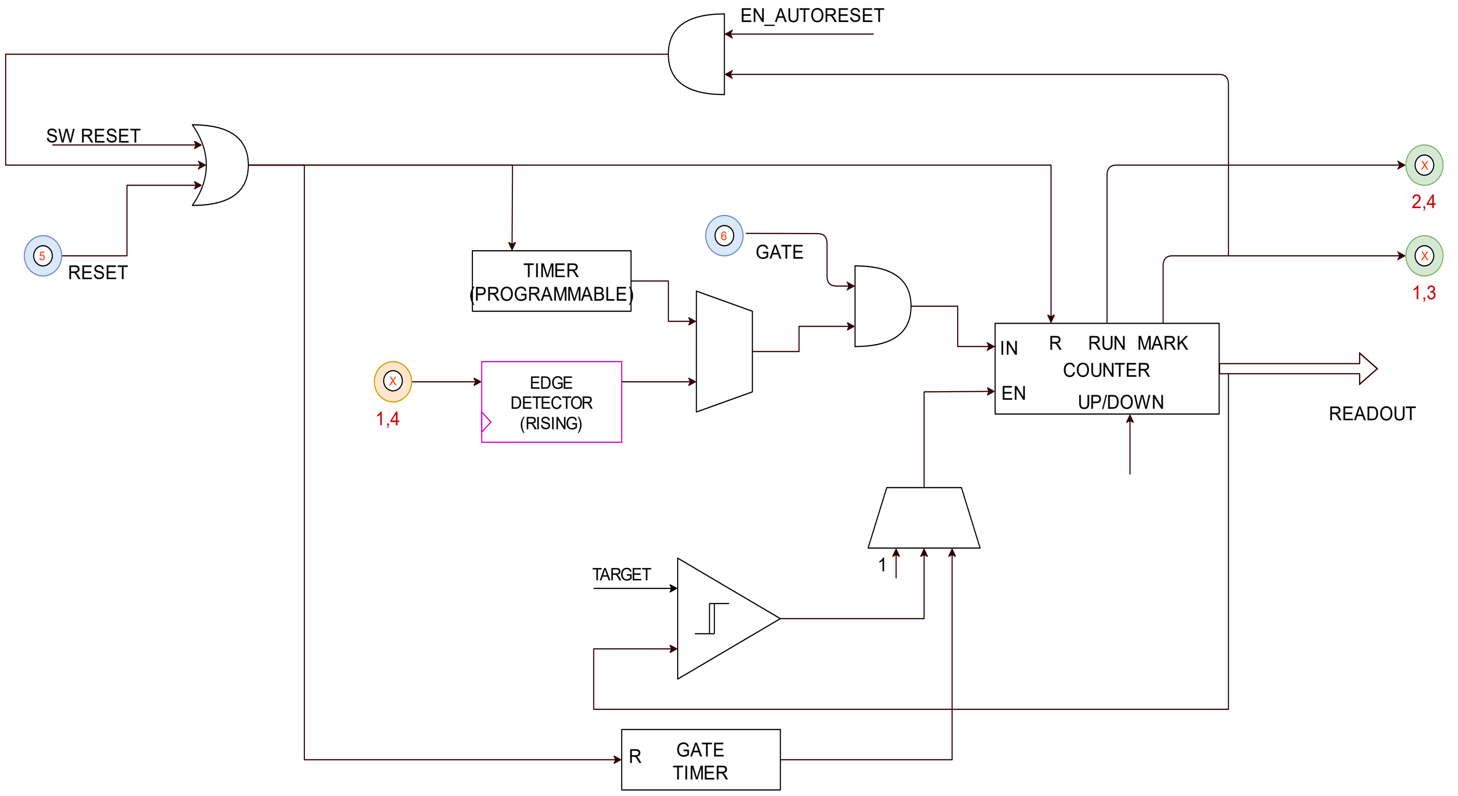
Counts first arrival in respect of a specific channel



* Counts total events and channel specific coincidence

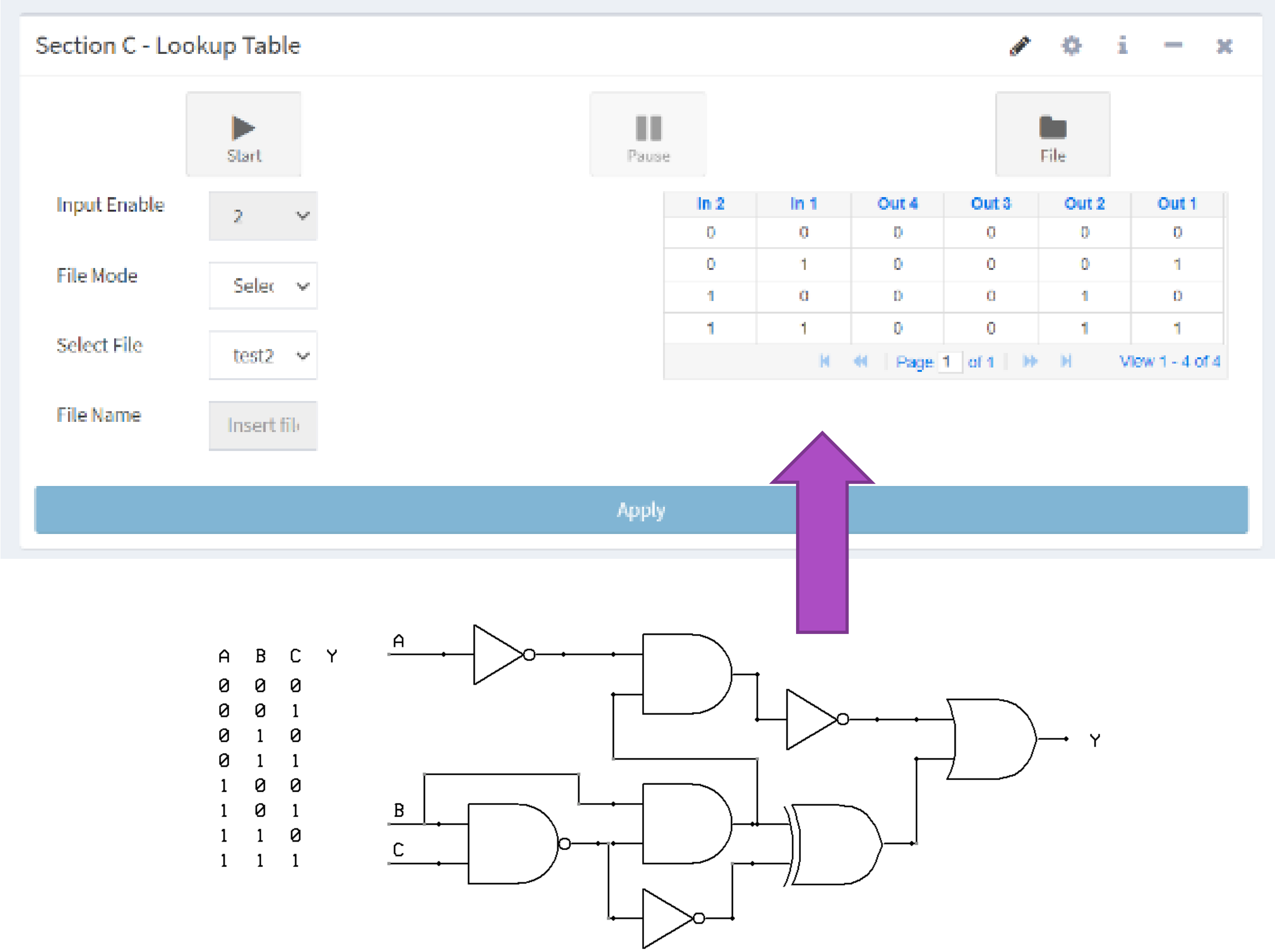
Timer

- **Up/down** time chronometer
- **Internal and external** time base (internal 100MHz)
- **Calibrated threshold** with +-5mV +-1% tolerance on the full range
- **Programmable target**
- **Run and Marker (preset)** output
- 64 bit depth
- Programmable auto-reset function
- Time-gated counter



Programmable LUT

- **Implement any** 6 input - 4 output **logic function** for each N1081B section
- **Programmable logic table** via web interface
- Load **ASCII file** generated with any logic synthesis tool
- Web based LUT editor



Time Tagging

- **Timestamp** up to 24 detector output with
- **10ns** resolution
- Up to 50KHz long time sustainable average rate per section, up to 20 MHz peak rate.
- **Integrated discriminator** allows to direct connected detector / pre to the device
- ASCII format file downloadable from web interface
- Acquisition time limited by PC available disk space
- Up to 4 channel per section
- T0 and GATE input

TIMESTAMP (ns)

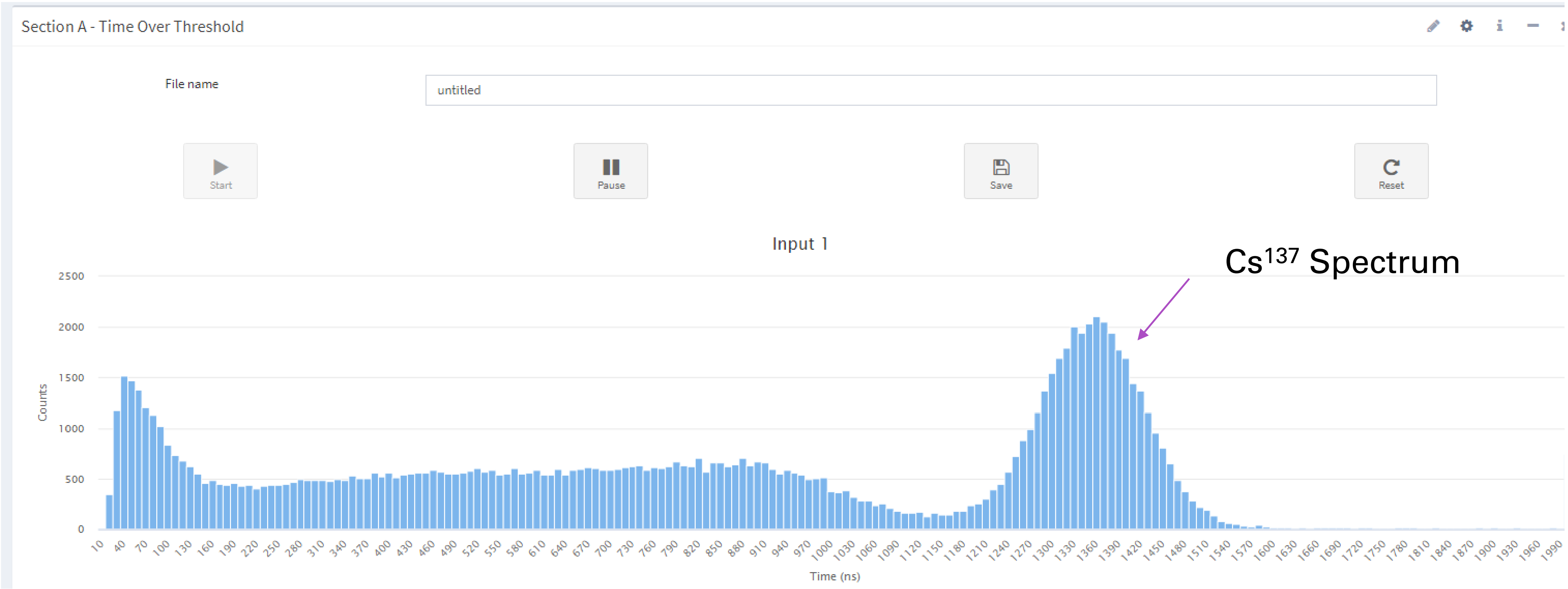
CHANNEL

	CHANNEL	TIMESTAMP (ns)
1	1	82527249248120
2	1	82527250248120
3	1	82527251248120
4	1	82527252248120
5	1	82527253248120
6	1	82527254248120
7	1	82527255248120
8	1	82527256248120
9	1	82527257248120
10	1	82527258248120
11	1	82527259248120
12	1	82527260248120

Time over Threshold

- **Bin size from 10ns to 1s**
- Up to **1024 bin**
- **Integrated discriminator** allows to direct connected detector / pre to the device
- **Online Spectrum** calculation
- ASCII format file downloadable from web interface

Time over Threshold

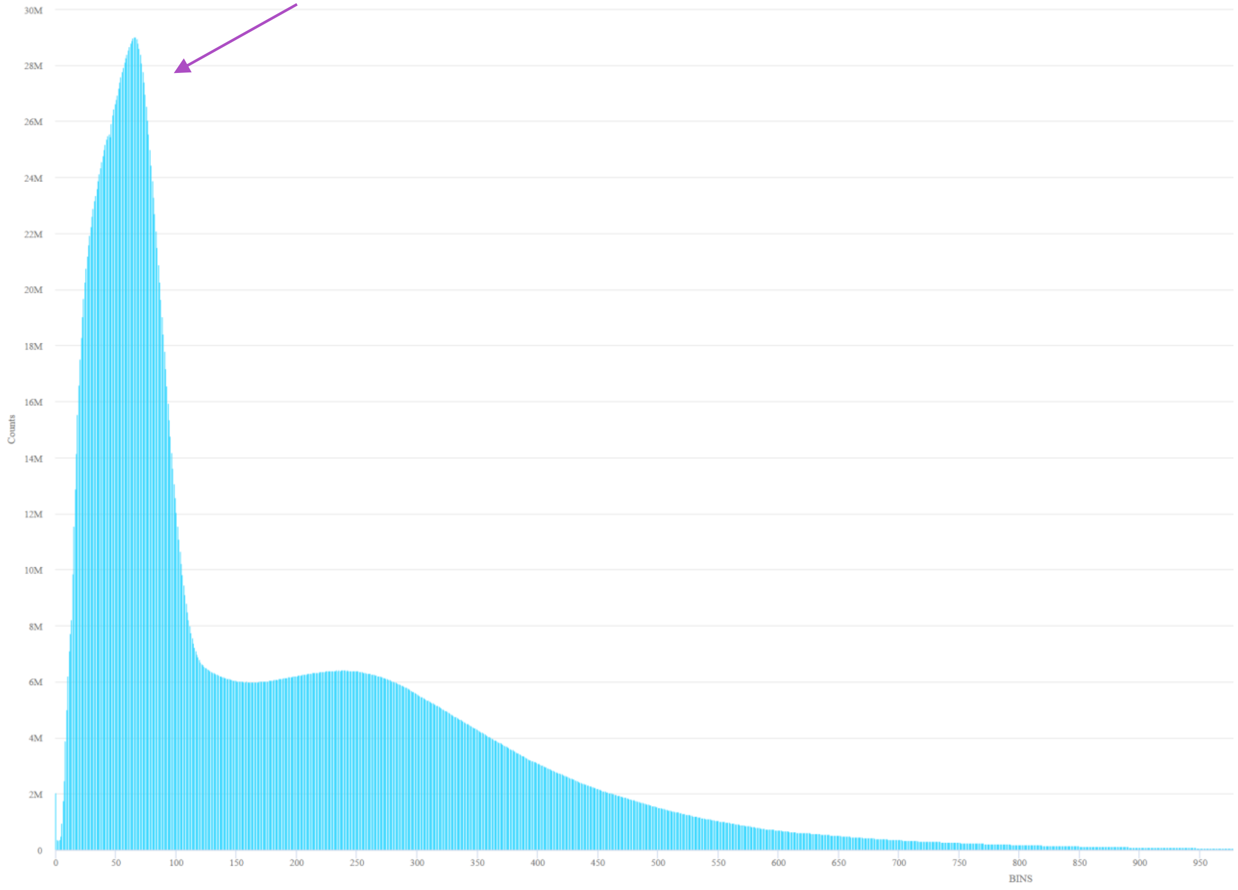


*measured by CAEN I-Spector device with 18x18x30mm Csi(TI) scintillator, 10ns bit size

Time of flight

- **Bin size from 10ns to 1s**
- Up to **1024 bin**
- **Integrated discriminator** allows to direct connected detector / pre to the device
- **Online Spectrum** calculation
- ASCII format file downloadable from web interface
- Independent programmable bin size

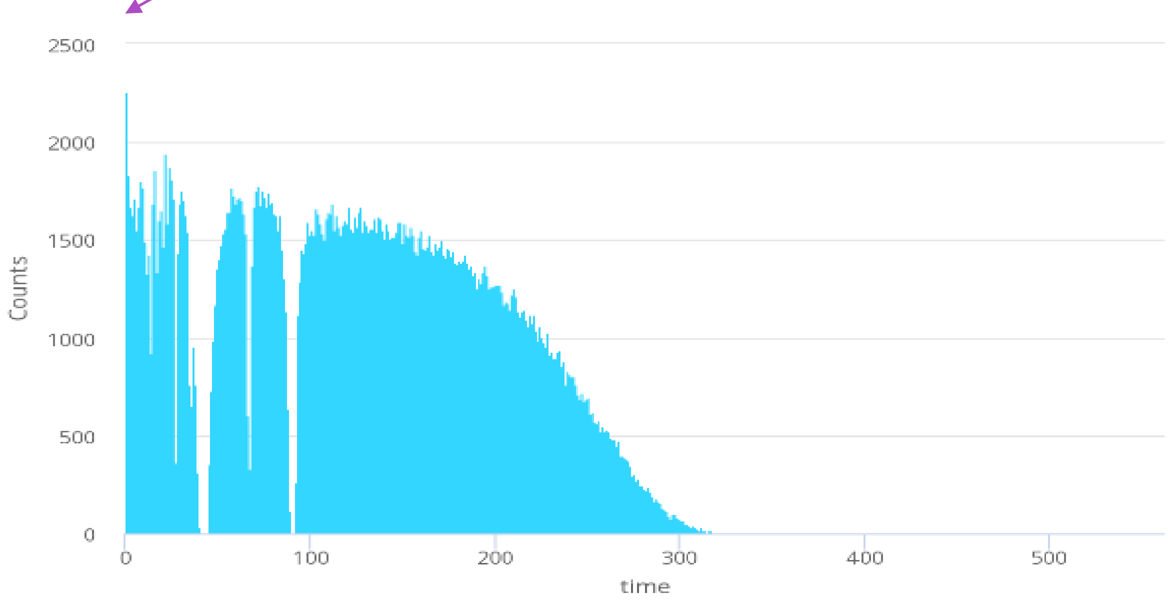
Neutron ToF on ISIS



Time of flight

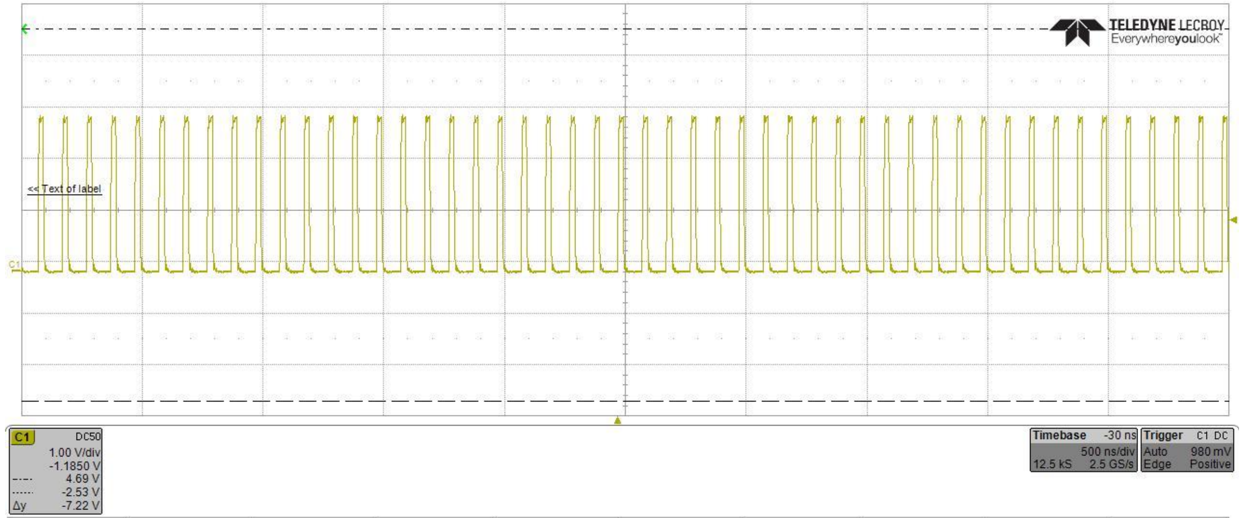
- **Bin size from 10ns to 1s**
- Up to **1024 bin**
- **Integrated discriminator** allows to direct connected detector / pre to the device
- **Online Spectrum** calculation
- ASCII format file downloadable from web interface
- Independent programmable bin size

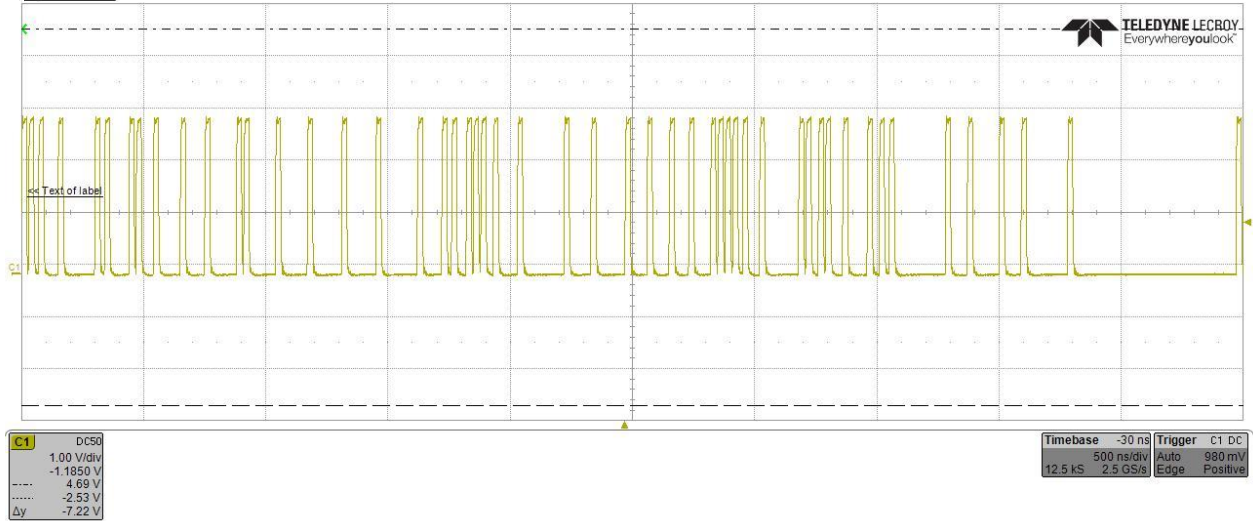
Cadmium ToF with clearly visible resonance



Generator

- **Embedded** programmable **signal generator**
- **Periodic and Poisson distributed** events
- 1Hz to 20MHz generation
- **10ns to 1s pulse width**
- **Low / High constant** generator

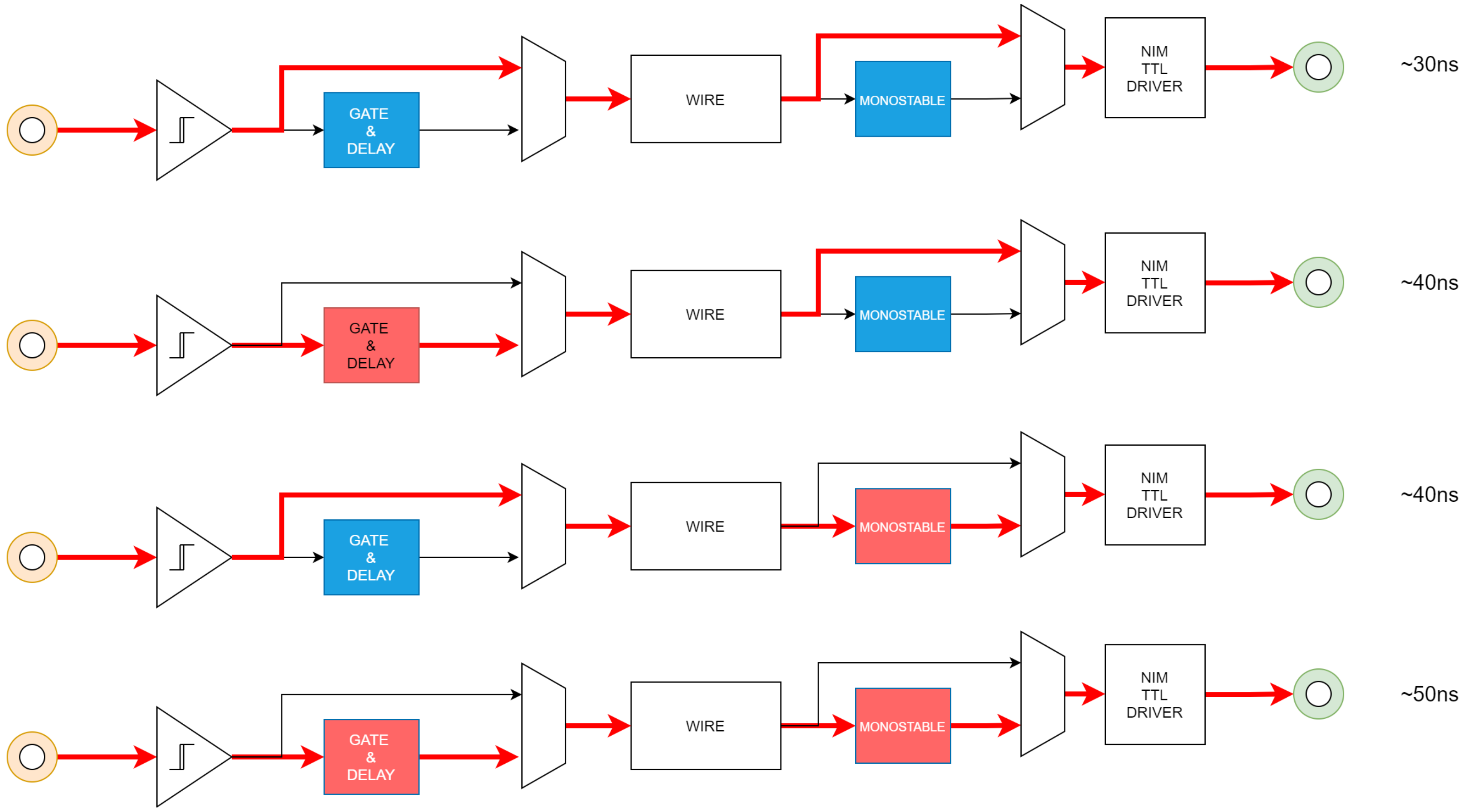




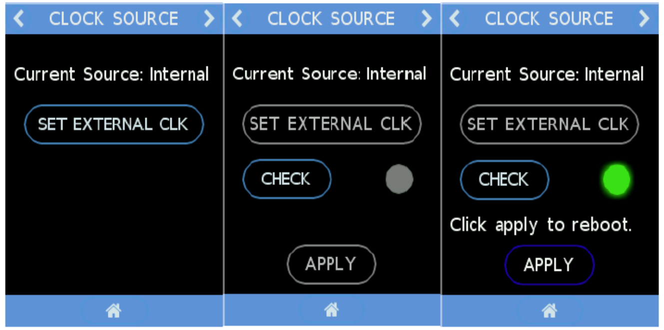
Section bypass

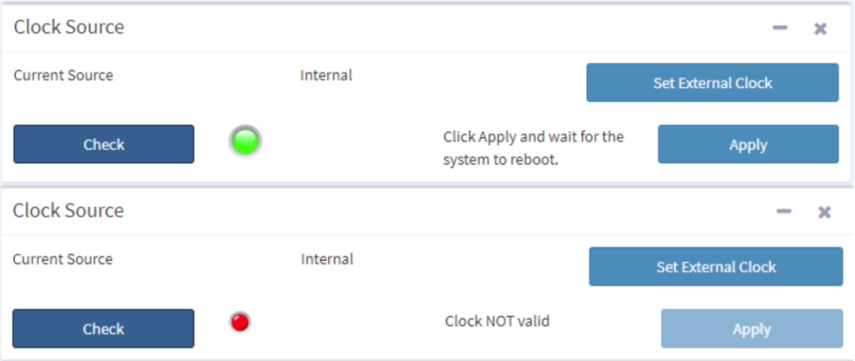
- **Daesy chain** section to extend the number of input of each function

Propagation delay



External clock



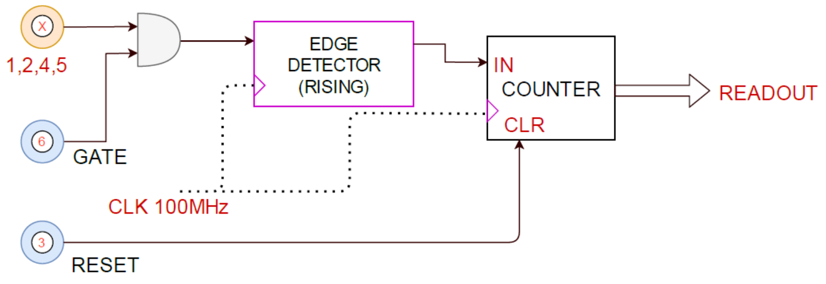


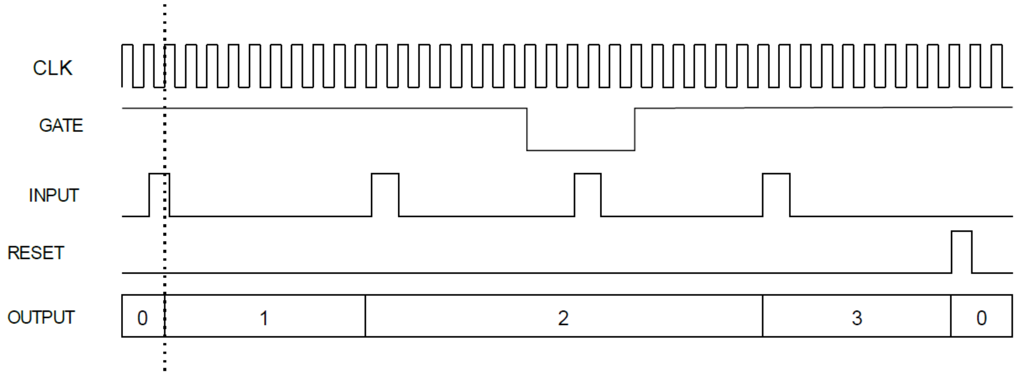
Localization in crowded crate

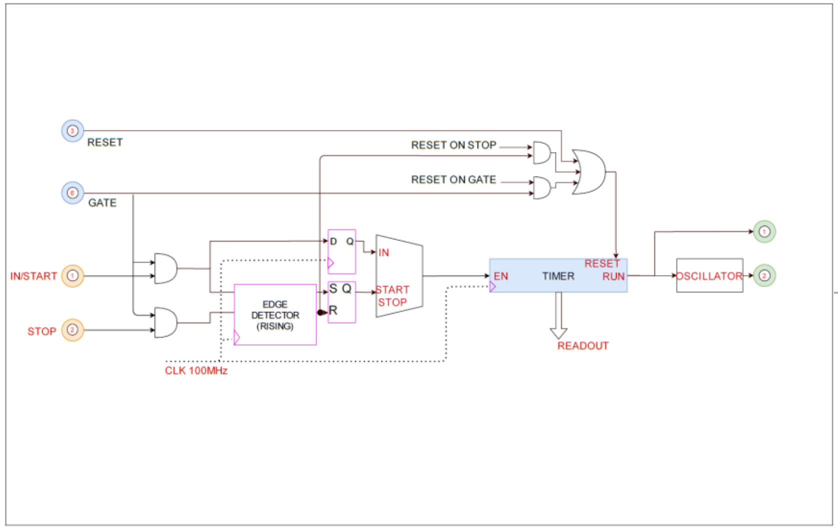
User Manual

COUNTER

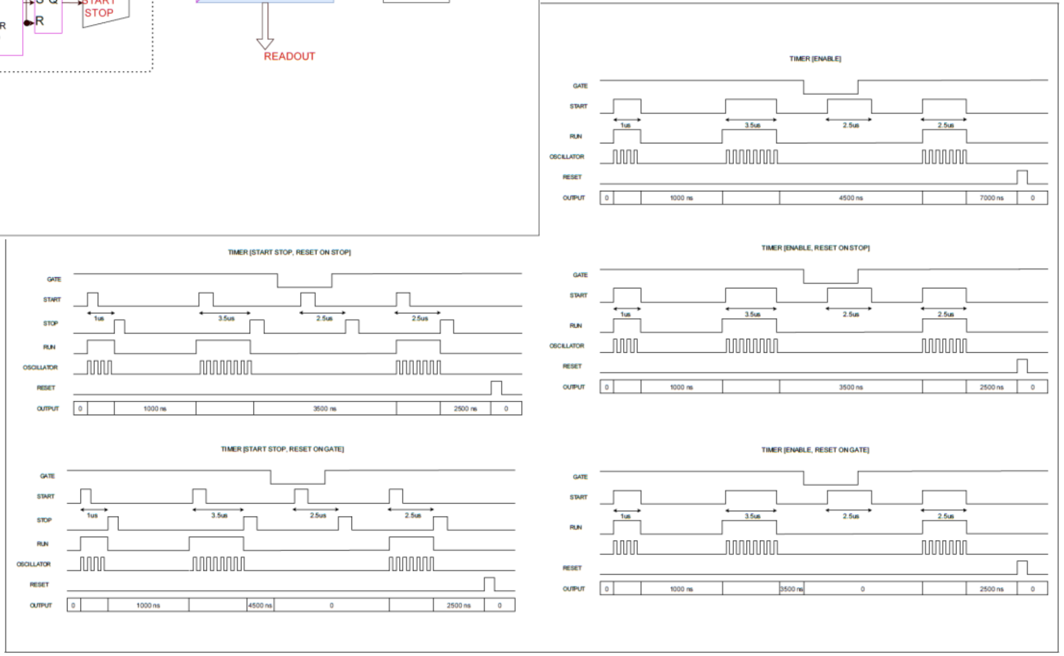
The COUNTER function counts the pulses on the enabled input channels. It is implemented as a synchronous function relying on a 100 MHz clock and it allows to count up to 40 MHz, with a depth of 64-bit, meaning that the counter never goes in overflow in practical situations. Input CH 3 is reserved for the reset signal: when it is logic-high signal, all counters are reset. Input CH 6 is reserved for the gate: if enabled, the input channels signals are counted only when the gate is logic-high.



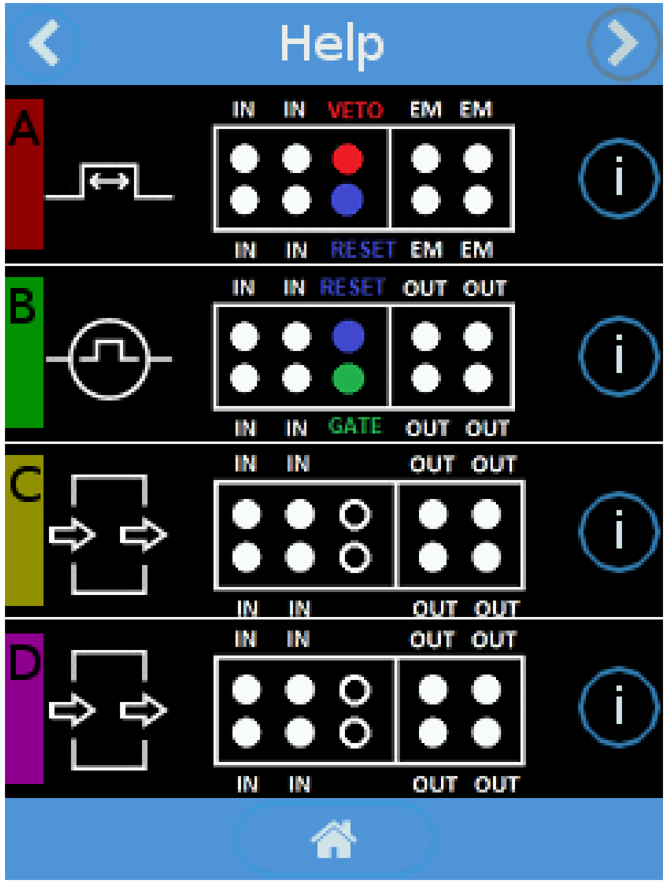


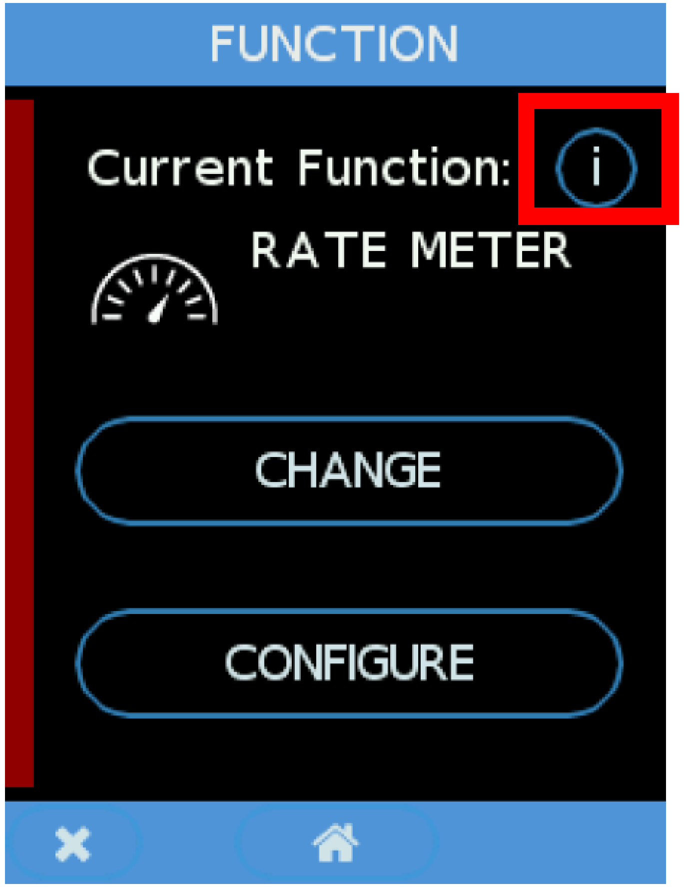


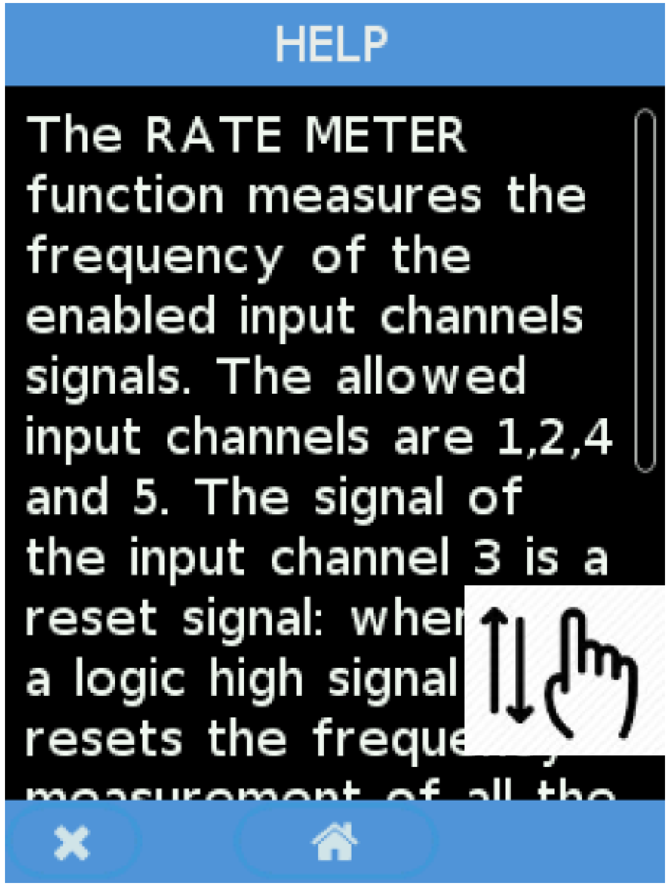
Timing diagram of every function is every possible configuration



Online help


Help
A
IN IN VETO EM EM
IN IN RESET EM EM
B
IN IN RESET OUT OUT
IN IN GATE OUT OUT
C
IN IN OUT OUT
IN IN OUT OUT
D
IN IN OUT OUT
IN IN OUT OUT


FUNCTION
Current Function:
RATE METER
CHANGE
CONFIGURE


HELP
The RATE METER function measures the frequency of the enabled input channels signals. The allowed input channels are 1,2,4 and 5. The signal of the input channel 3 is a reset signal: when a logic high signal resets the freque

Configuration Memory

Thank you for your attention

Any question/curiosity?